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Gravitation

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 201 questions

Q1. JEE Main 2026 (Online) 5th April Evening Shift

A body of mass m is taken from the surface of earth to a height equal to twice the radius of earth (R_e). The increase in potential energy will be ____ .

(g is acceleration due to gravity at the surface of earth)

(A) $\frac{1}{2}mgR_e$ (B) $\frac{3}{4}mgR_e$ (C) $\frac{1}{4}mgR_e$ (D) $\frac{2}{3}mgR_e$

Open solution

Q2. JEE Main 2026 (Online) 5th April Morning Shift

When one moves from a point 16 km below the earth's surface to a point 16 km above the earth's surface, the change in g is approximately $\alpha\%$. The value of α is ____ .

(Take radius of the earth $= 6400\ \text{km}$.)

(A) 0.12 (B) 0.25 (C) 0.50 (D) 0.75

Open solution

Q3. JEE Main 2026 (Online) 4th April Evening Shift

The height in terms of radius of the earth (R), at which the acceleration due to gravity becomes $\frac{g}{9}$, where g is acceleration due to gravity on earth's surface, is

____ .

(A) $\sqrt{3}R$ (B) $2\sqrt{2}R$ (C) $2R$ (D) $\frac{4}{9}R$

Open solution

Q4. JEE Main 2026 (Online) 2nd April Evening Shift

If a body of mass 1 kg falls on the earth from infinity, it attains velocity (v) and kinetic energy (k) on reaching the surface of earth. The values of v and k respectively are _____.

(Take radius of earth to be 6400 km and g = 9.8 m/s^2)

(A) 11.2 km/s; 6.27×10^7 J (B) 11.2 km/s; 12.54×10^7 J

(C) 8.8 km/s; 6.27×10^7 J (D) 8.8 km/s; 12.54×10^7 J

Open solution

Q5. JEE Main 2026 (Online) 2nd April Morning Shift

A planet (P_1) is moving around the star of mass $2M$ in the orbit of radius R. Another planet (P_2) is moving around another star of mass $4M$ in a orbit of radius $2R$. Ratio of time periods of revolution of P_2 and P_1 is _____.

(A) $\frac{1}{2}$ (B) 2 (C) 4 (D) $\frac{1}{4}$

Open solution

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Q6. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Three masses $200\ \text{kg}, 300\ \text{kg}$ and 400 kg are placed at the vertices of an equilateral triangle with sides 20 m . They are rearranged on the vertices of a bigger triangle of side 25 m and with the same centre. The work done in this process ____ J. (Gravitational constant $\text{G} = 6.7 \times 10^{-11}\ \text{N m}^2/\text{kg}^2$)

(A) 4.77×10^{-7} (B) 1.74×10^{-7} (C) 9.86×10^{-6} (D) 2.85×10^{-7}

Q7. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Given below are two statements :

Statement I : A satellite is moving around earth in the orbit very close to the earth surface. The time period of revolution of satellite depends upon the density of earth.

Statement II : The time period of revolution of the satellite is $T = 2\pi\sqrt{\frac{R_e}{g}}$ (for satellite very close to the earth surface), where R_e radius of earth and g acceleration due to gravity. In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q8. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

The escape velocity from a spherical planet A is $10\ \text{km/s}$. The escape velocity from another planet B whose density and radius are 10% of those of planet A, is ____ m/s.

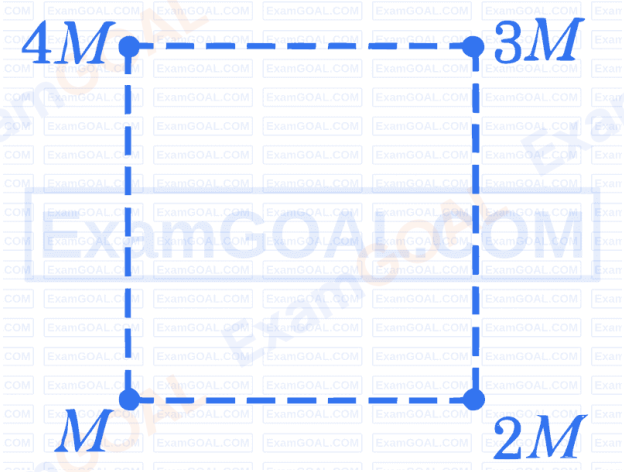
(A) 1000 (B) $200\sqrt{5}$ (C) $1000\sqrt{2}$ (D) $100\sqrt{10}$

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Q9. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Net gravitational force at the center of a square is found to be F_1 when four particles having mass $M, 2M, 3M$ and $4M$ are placed at the four corners of the square as shown in figure and it is F_2 when the positions of $3M$ and $4M$ are interchanged. The ratio $\frac{F_1}{F_2}$ is $\frac{\alpha}{\sqrt{5}}$. The value of α is ____ .



(A) 2 (B) $2\sqrt{5}$ (C) 1 (D) 3

Q10. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Initially a satellite of 100 kg is in a circular orbit of radius $1.5\mathrm{R_E}$. This satellite can be moved to a circular orbit of radius $3R_E$ by supplying $\alpha \times 10^6$ J of energy. The value of α is ____ .

(Take Radius of Earth $R_E = 6 \times 10^6$ m and $\mathrm{g} = 10\,\mathrm{m/s^2}$)

(A) 500 (B) 1000 (C) 100 (D) 150

Q11. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The radius vector from the Sun to a planet sweeps out equal areas in equal intervals of time and thus areal velocity of planet is constant.

Reason (R): For a central force field the angular momentum is a constant.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) (A) is not correct but (R) is correct

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(C) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(D) (A) is correct but (R) is not correct

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Q12. JEE Main 2025 (Online) 4th April Evening Shift

An object is kept at rest at a distance of $3R$ above the earth's surface where R is earth's radius. The minimum speed with which it must be projected so that it does not return to earth is : (Assume $\mathrm{M}=$ mass of earth, $\mathrm{G}=$ Universal gravitational constant)

Open solution

(A) $\sqrt{\frac{3\mathrm{GM}}{\mathrm{R}}}$

(B) $\sqrt{\frac{2\mathrm{GM}}{\mathrm{R}}}$

(C) $\sqrt{\frac{\mathrm{GM}}{2\mathrm{R}}}$

(D) $\sqrt{\frac{\mathrm{GM}}{\mathrm{R}}}$

Q13. JEE Main 2025 (Online) 4th April Morning Shift

Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**

Open solution

Assertion A : The kinetic energy needed to project a body of mass m from earth surface to infinity is $\frac{1}{2}\mathrm{mgR}$, where R is the radius of earth.

Reason R : The maximum potential energy of a body is zero when it is projected to infinity from earth surface.

In the light of the above statements, choose the correct answer from the options given below

(A) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Q14. JEE Main 2025 (Online) 28th January Evening Shift

Earth has mass 8 times and radius 2 times that of a planet. If the escape velocity from the earth is 11.2 km/s, the escape velocity in km/s from the planet will be:

Open solution

(A) 8.4 (B) 11.2 (C) 5.6 (D) 2.8

Q15. JEE Main 2025 (Online) 24th January Morning Shift

A satellite is launched into a circular orbit of radius ' R ' around the earth. A second satellite is launched into an orbit of radius 1.03 R . The time period of revolution of the second satellite is larger than the first one approximately by

Open solution

(A) 3% (B) 2.5% (C) 4.5% (D) 9%

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Q16. JEE Main 2025 (Online) 23rd January Evening Shift

If a satellite orbiting the Earth is 9 times closer to the Earth than the Moon, what is the time period of rotation of the satellite? Given rotational time period of Moon $= 27$ days and gravitational attraction between the satellite and the moon is neglected.

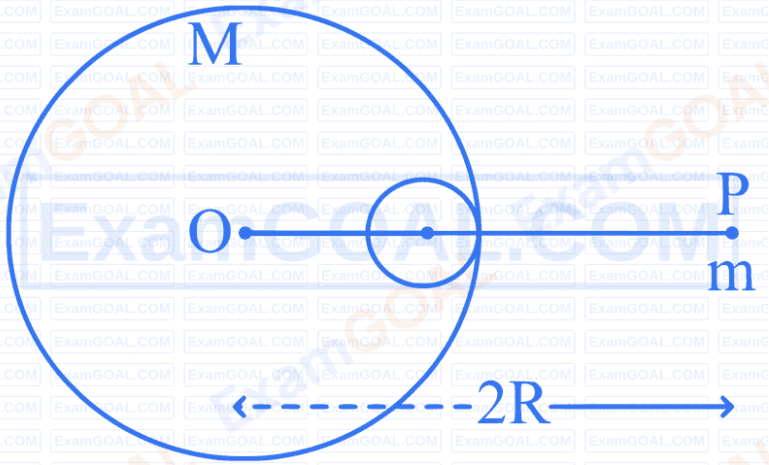
Open solution

(A) 3 days (B) 27 days (C) 81 days (D) 1 day

Q17. JEE Main 2025 (Online) 22nd January Morning Shift

A small point of mass m is placed at a distance $2R$ from the centre ' O ' of a big uniform solid sphere of mass M and radius R . The gravitational force on ' m ' due to M is F_1. A spherical part of radius $\mathrm{R}/3$ is removed from the big sphere as shown in the figure and the gravitational force on m due to remaining part of M is found to be F_2. The value of ratio $F_1 : F_2$ is

Open solution



(A) 11 : 10 (B) 12 : 11 (C) 16 : 9 (D) 12 : 9

Q18. JEE Main 2024 (Online) 9th April Evening Shift

A satellite of 10^3 kg mass is revolving in circular orbit of radius $2R$. If $\frac{10^4 R}{6}$ J energy is supplied to the satellite, it would revolve in a new circular orbit of radius

(use $g = 10\ \mathrm{m/s^2}$, $R =$ radius of earth)

Open solution

(A) 4 R (B) 6 R (C) 2.5 R (D) 3 R

Q19. JEE Main 2024 (Online) 9th April Morning Shift

An astronaut takes a ball of mass m from earth to space. He throws the ball into a circular orbit about earth at an altitude of 318.5 km. From earth's surface to the orbit, the change in total mechanical energy of the ball is $x\frac{\mathrm{GM_e m}}{21\mathrm{R_e}}$. The value of x is (take $\mathrm{R_e} = 6370$ km) :

Open solution

(A) 12 (B) 11 (C) 9 (D) 10

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Q20. JEE Main 2024 (Online) 8th April Evening Shift

Two satellite A and B go round a planet in circular orbits having radii 4R and R respectively. If the speed of A is $3v$, the speed of B will be :

(A) $6v$ (B) $\frac{4}{3}v$ (C) $3v$ (D) $12v$

Open solution

Q21. JEE Main 2024 (Online) 8th April Morning Shift

Two planets A and B having masses m_1 and m_2 move around the sun in circular orbits of r_1 and r_2 radii respectively. If angular momentum of A is L and that of B is $3\,\mathrm{L}$, the ratio of time period $\left(\frac{T_A}{T_B}\right)$ is:

(A) $\left(\frac{r_2}{r_1}\right)^{\frac{3}{2}}$ (B) $27\left(\frac{m_1}{m_2}\right)^3$

(C) $\left(\frac{r_1}{r_2}\right)^3$ (D) $\frac{1}{27}\left(\frac{m_2}{m_1}\right)^3$

Open solution

Q22. JEE Main 2024 (Online) 6th April Evening Shift

Assuming the earth to be a sphere of uniform mass density, a body weighed $300\,\mathrm{N}$ on the surface of earth. How much it would weigh at R/4 depth under surface of earth ?

(A) 75 N (B) 375 N (C) 300 N (D) 225 N

Open solution

Q23. JEE Main 2024 (Online) 6th April Morning Shift

To project a body of mass m from earth's surface to infinity, the required kinetic energy is (assume, the radius of earth is $R_E, g =$ acceleration due to gravity on the surface of earth):

(A) $1/2mgR_E$ (B) $4mgR_E$ (C) mgR_E (D) $2mgR_E$

Open solution

Q24. JEE Main 2024 (Online) 5th April Evening Shift

A satellite revolving around a planet in stationary orbit has time period 6 hours. The mass of planet is one-fourth the mass of earth. The radius of orbit of the satellite is :

(Given = Radius of geo-stationary orbit for earth is $4.2 \times 10^4\,\mathrm{km}$)

(A) $1.68 \times 10^5\,\mathrm{km}$ (B) $1.4 \times 10^4\,\mathrm{km}$ (C) $8.4 \times 10^4\,\mathrm{km}$ (D) $1.05 \times 10^4\,\mathrm{km}$

Open solution

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Q25. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

Match List I with List II :

	LIST I		LIST II
A.	Kinetic energy of planet	I.	$-GMm/a$
B.	Gravitation Potential energy of sun-planet system	II.	$GMm/2a$
C.	Total mechanical energy of planet	III.	$\frac{Gm}{r}$
D.	Escape energy at the surface of planet for unit mass object	IV.	$-GMm/2a$

(Where a = radius of planet orbit, r = radius of planet, M = mass of Sun, m = mass of planet)

Choose the correct answer from the options given below :

(A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(B) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(D) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

Q26. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

A $90\ kg$ body placed at $2R$ distance from surface of earth experiences gravitational pull of :

(R = Radius of earth, $g = 10\ m\ s^{-2}$)

(A) 300 N

(B) 225 N

(C) 100 N

(D) 120 N

Q27. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

Correct formula for height of a satellite from earth's surface is :

(A) $\left(\frac{T^2R^2g}{4\pi^2}\right)^{1/3} - R$

(B) $\left(\frac{T^2R^2g}{4\pi}\right)^{1/2} - R$

(C) $\left(\frac{T^2R^2g}{4\pi^2}\right)^{-1/3} + R$

(D) $\left(\frac{T^2R^2}{4\pi^2g}\right)^{1/3} - R$

Q28. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

A metal wire of uniform mass density having length L and mass M is bent to form a semicircular arc and a particle of mass m is placed at the centre of the arc. The gravitational force on the particle by the wire is :

(A) $\frac{GmM\pi^2}{L^2}$

(B) $\frac{GMm\pi}{2L^2}$

(C) 0

(D) $\frac{2GmM\pi}{L^2}$

Q29. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

A light planet is revolving around a massive star in a circular orbit of radius R with a period of revolution T. If the force of attraction between planet and star is proportional to $R^{-3/2}$ then choose the correct option :

(A) $T^2 \propto R^{7/2}$

(B) $T^2 \propto R^3$

(C) $T^2 \propto R^{5/2}$

(D) $T^2 \propto R^{3/2}$

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Q30. JEE Main 2024 (Online) 1st February Morning Shift

If R is the radius of the earth and the acceleration due to gravity on the surface of earth is $g = \pi^2\ \mathrm{m/s^2}$, then the length of the second's pendulum at a height $\mathrm{h} = 2R$ from the surface of earth will be :

Open solution

(A) $\frac{1}{9}\ \mathrm{m}$ (B) $\frac{8}{9}\ \mathrm{m}$ (C) $\frac{2}{9}\ \mathrm{m}$ (D) $\frac{4}{9}\ \mathrm{m}$

Q31. JEE Main 2024 (Online) 31st January Evening Shift

The mass of the moon is $\frac{1}{144}$ times the mass of a planet and its diameter is $\frac{1}{16}$ times the diameter of a planet. If the escape velocity on the planet is v, the escape velocity on the moon will be :

Open solution

(A) $\frac{\mathrm{v}}{4}$ (B) $\frac{\mathrm{v}}{6}$ (C) $\frac{\mathrm{v}}{12}$ (D) $\frac{\mathrm{v}}{3}$

Q32. JEE Main 2024 (Online) 31st January Morning Shift

Four identical particles of mass m are kept at the four corners of a square. If the gravitational force exerted on one of the masses by the other masses is $\left(\frac{2\sqrt{2}+1}{32}\right)\frac{\mathrm{Gm}^2}{L^2}$, the length of the sides of the square is

Open solution

(A) 4L (B) 3L (C) 2L (D) $\frac{L}{2}$

Q33. JEE Main 2024 (Online) 30th January Evening Shift

Escape velocity of a body from earth is $11.2\ \mathrm{km/s}$. If the radius of a planet be one-third the radius of earth and mass be one-sixth that of earth, the escape velocity from the planet is :

Open solution

(A) 7.9 km/s (B) 8.4 km/s (C) 4.2 km/s (D) 11.2 km/s

Q34. JEE Main 2024 (Online) 30th January Morning Shift

The gravitational potential at a point above the surface of earth is $-5.12 \times 10^7\ \mathrm{J/kg}$ and the acceleration due to gravity at that point is $6.4\ \mathrm{m/s^2}$. Assume that the mean radius of earth to be $6400\ \mathrm{km}$. The height of this point above the earth's surface is :

Open solution

(A) 1600 km (B) 1200 km (C) 540 km (D) 1000 km

Q35. JEE Main 2024 (Online) 29th January Evening Shift

A planet takes 200 days to complete one revolution around the Sun. If the distance of the planet from Sun is reduced to one fourth of the original distance, how many days will it take to complete one revolution :

Open solution

(A) 20 (B) 50 (C) 100 (D) 25

Q36. JEE Main 2024 (Online) 29th January Morning Shift

At what distance above and below the surface of the earth a body will have same weight. (take radius of earth as R.)

Open solution

(A) $\frac{\sqrt{3}R-R}{2}$ (B) $\frac{R}{2}$ (C) $\frac{\sqrt{5}R-R}{2}$ (D) $\sqrt{5}R - R$

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Q37. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The angular speed of the moon in its orbit about the earth is more than the angular speed of the earth in its orbit about the sun.

Reason (R) : The moon takes less time to move around the earth than the time taken by the earth to move around the sun.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(B) (A) is correct but (R) is not correct

(C) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(D) (A) is not correct but (R) is correct

Q38. JEE Main 2024 (Online) 27th January Morning Shift

Open solution

The acceleration due to gravity on the surface of earth is g. If the diameter of earth reduces to half of its original value and mass remains constant, then acceleration due to gravity on the surface of earth would be :

(A) g/4 **(B)** 2g **(C)** g/2 **(D)** 4g

Q39. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

Two identical particles each of mass ' m ' go round a circle of radius a under the action of their mutual gravitational attraction. The angular speed of each particle will be :

(A) $\sqrt{\frac{Gm}{2a^3}}$ **(B)** $\sqrt{\frac{Gm}{a^3}}$ **(C)** $\sqrt{\frac{Gm}{8a^3}}$ **(D)** $\sqrt{\frac{Gm}{4a^3}}$

Q40. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

A body is released from a height equal to the radius (R) of the earth. The velocity of the body when it strikes the surface of the earth will be

(Given $g =$ acceleration due to gravity on the earth.)

(A) $\sqrt{\frac{gR}{2}}$ **(B)** $\sqrt{4gR}$ **(C)** $\sqrt{2gR}$ **(D)** $\sqrt{gR}$

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Q41. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

Given below are two statements:

Statement I : For a planet, if the ratio of mass of the planet to its radius increases, the escape velocity from the planet also increases.

Statement II : Escape velocity is independent of the radius of the planet.

In the light of above statements, choose the most appropriate answer form the options given below

(A) Both Statement I and Statement II are correct

(B) Statement I is correct but statement II is incorrect

(C) Both Statement I and Statement II are incorrect

(D) Statement I is incorrect but statement II is correct

Q42. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

Two planets A and B of radii R and 1.5 R have densities ρ and $\rho/2$ respectively. The ratio of acceleration due to gravity at the surface of B to A is:

(A) 2 : 1 (B) 2 : 3 (C) 4 : 3 (D) 3 : 4

Q43. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

A planet having mass $9\mathrm{Me}$ and radius $4\mathrm{R_e}$, where Me and Re are mass and radius of earth respectively, has escape velocity in $\mathrm{km/s}$ given by:

(Given escape velocity on earth $\mathrm{V_e} = 11.2 \times 10^3\ \mathrm{m/s}$)

(A) 33.6 (B) 11.2 (C) 16.8 (D) 67.2

Q44. JEE Main 2023 (Online) 12th April Morning Shift

Open solution

The ratio of escape velocity of a planet to the escape velocity of earth will be:-

Given: Mass of the planet is 16 times mass of earth and radius of the planet is 4 times the radius of earth.

(A) $1:4$ (B) $1:\sqrt{2}$ (C) $4:1$ (D) $2:1$

Q45. JEE Main 2023 (Online) 12th April Morning Shift

Open solution

Two satellites A and B move round the earth in the same orbit. The mass of A is twice the mass of B. The quantity which is same for the two satellites will be

(A) Potential energy (B) Kinetic energy (C) Total energy (D) Speed

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Q46. JEE Main 2023 (Online) 11th April Evening Shift

A space ship of mass $2 \times 10^4\ \mathrm{kg}$ is launched into a circular orbit close to the earth surface. The additional velocity to be imparted to the space ship in the orbit to overcome the gravitational pull will be (if $g = 10\ \mathrm{m/s^2}$ and radius of earth $= 6400\ \mathrm{km}$):

(A) $7.9(\sqrt{2}-1)\mathrm{km/s}$ (B) $11.2(\sqrt{2}-1)\mathrm{km/s}$

(C) $7.4(\sqrt{2}-1)\mathrm{km/s}$ (D) $8(\sqrt{2}-1)\mathrm{km/s}$

Open solution

Q47. JEE Main 2023 (Online) 11th April Evening Shift

If V is the gravitational potential due to sphere of uniform density on its surface, then its value at the center of sphere will be:-

(A) $\frac{3\,\mathrm{V}}{2}$ (B) $\frac{\mathrm{V}}{2}$ (C) $\frac{4}{3}\,\mathrm{V}$ (D) V

Open solution

Q48. JEE Main 2023 (Online) 11th April Morning Shift

The radii of two planets 'A' and 'B' are 'R' and '4R' and their densities are ρ and $\rho/3$ respectively. The ratio of acceleration due to gravity at their surfaces $(g_A : g_B)$ will be:

(A) 3 : 16 (B) 4 : 3 (C) 1 : 16 (D) 3 : 4

Open solution

Q49. JEE Main 2023 (Online) 10th April Evening Shift

The time period of a satellite, revolving above earth's surface at a height equal to R will be

(Given $g = \pi^2\ \mathrm{m/s^2}, \mathrm{R} =$ radius of earth)

(A) $\sqrt{32R}$ (B) $\sqrt{4\mathrm{R}}$ (C) $\sqrt{8R}$ (D) $\sqrt{2R}$

Open solution

Q50. JEE Main 2023 (Online) 10th April Evening Shift

Given below are two statements:

Statement I : Rotation of the earth shows effect on the value of acceleration due to gravity (g)

Statement II : The effect of rotation of the earth on the value of 'g' at the equator is minimum and that at the pole is maximum.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but statement II is true

(B) Statement I is true but statement II is false

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Open solution

Q51. JEE Main 2023 (Online) 10th April Morning Shift

Two satellites of masses m and 3m revolve around the earth in circular orbits of radii r & 3r respectively. The ratio of orbital speeds of the satellites respectively is

(A) 3 : 1 (B) $\sqrt{3}$: 1 (C) 1 : 1 (D) 9 : 1

Open solution

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Q52. JEE Main 2023 (Online) 10th April Morning Shift

Assuming the earth to be a sphere of uniform mass density, the weight of a body at a depth $d = \frac{R}{2}$ from the surface of earth, if its weight on the surface of earth is 200 N, will be:

(Given R = radius of earth)

(A) 100 N (B) 400 N (C) 300 N (D) 500 N

Open solution

Q53. JEE Main 2023 (Online) 8th April Evening Shift

The acceleration due to gravity at height h above the earth if $h << \mathrm{R}$ (Radius of earth) is given by

(A) $g' = g\left(1 - \frac{2h}{R}\right)$ (B) $g' = g\left(1 - \frac{2h^2}{R^2}\right)$

(C) $g' = g\left(1 - \frac{h^2}{2R^2}\right)$ (D) $g' = g\left(1 - \frac{h}{2R}\right)$

Open solution

Q54. JEE Main 2023 (Online) 8th April Evening Shift

The orbital angular momentum of a satellite is L, when it is revolving in a circular orbit at height h from earth surface. If the distance of satellite from the earth centre is increased by eight times to its initial value, then the new angular momentum will be -

(A) 9L (B) 8L (C) 4L (D) 3L

Open solution

Q55. JEE Main 2023 (Online) 8th April Morning Shift

Given below are two statements:

Statement I: If E be the total energy of a satellite moving around the earth, then its potential energy will be $\frac{E}{2}$.

Statement II: The kinetic energy of a satellite revolving in an orbit is equal to the half the magnitude of total energy E.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Both Statement I and Statement II are incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Statement I is correct but Statement II is incorrect

(D) Both Statement I and Statement II are correct

Open solution

Q56. JEE Main 2023 (Online) 8th April Morning Shift

The weight of a body on the earth is $400\ \mathrm{N}$. Then weight of the body when taken to a depth half of the radius of the earth will be:

(A) 300 N (B) 200 N (C) 100 N (D) Zero

Open solution

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Q57. JEE Main 2023 (Online) 6th April Evening Shift

The weight of a body on the surface of the earth is $100\ \mathrm{N}$. The gravitational force on it when taken at a height, from the surface of earth, equal to one-fourth the radius of the earth is:

(A) 50 N (B) 64 N (C) 25 N (D) 100 N

Open solution

Q58. JEE Main 2023 (Online) 6th April Evening Shift

Choose the incorrect statement from the following:

(A) The linear speed of a planet revolving around the sun remains constant.

(B) When a body falls towards earth, the displacement of earth towards the body is negligible.

(C) The speed of satellite in a given circular orbit remains constant.

(D) For a planet revolving around the sun in an elliptical orbit, the total energy of the planet remains constant.

Open solution

Q59. JEE Main 2023 (Online) 6th April Morning Shift

Given below are two statements : one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$.

Assertion A : Earth has atmosphere whereas moon doesn't have any atmosphere.

Reason R : The escape velocity on moon is very small as compared to that on earth.

In the light of the above statements, choose the correct answer from the options given below:

(A) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(D) $\mathbf{A}$ is true but $\mathbf{R}$ is false

Open solution

Q60. JEE Main 2023 (Online) 6th April Morning Shift

A planet has double the mass of the earth. Its average density is equal to that of the earth. An object weighing W on earth will weigh on that planet:

(A) $2^{2/3}\ \mathrm{W}$ (B) W (C) $2\ \mathrm{W}$ (D) $2^{1/3}\ \mathrm{W}$

Open solution

Q61. JEE Main 2023 (Online) 1st February Evening Shift

The escape velocities of two planets A and B are in the ratio $1:2$. If the ratio of their radii respectively is $1:3$, then the ratio of acceleration due to gravity of planet A to the acceleration of gravity of planet B will be :

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{3}{2}$

Open solution

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Q62. JEE Main 2023 (Online) 1st February Morning Shift

If earth has a mass nine times and radius twice to that of a planet P. Then $\frac{v_e}{3}\sqrt{x}\,\mathrm{ms}^{-1}$ will be the minimum velocity required by a rocket to pull out of gravitational force of P, where v_e is escape velocity on earth. The value of x is

Open solution

(A) 1 (B) 3 (C) 2 (D) 18

Q63. JEE Main 2023 (Online) 1st February Morning Shift

Given below are two statements:

Statement I: Acceleration due to gravity is different at different places on the surface of earth.

Statement II: Acceleration due to gravity increases as we go down below the earth's surface.

In the light of the above statements, choose the correct answer from the options given below

Open solution

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q64. JEE Main 2023 (Online) 31st January Evening Shift

A body weight W, is projected vertically upwards from earth's surface to reach a height above the earth which is equal to nine times the radius of earth. The weight of the body at that height will be :

Open solution

(A) $\frac{W}{91}$ (B) $\frac{\mathrm{W}}{3}$ (C) $\frac{\mathrm{W}}{100}$ (D) $\frac{\mathrm{W}}{9}$

Q65. JEE Main 2023 (Online) 31st January Morning Shift

At a certain depth "d " below surface of earth, value of acceleration due to gravity becomes four times that of its value at a height $3\mathrm{R}$ above earth surface. Where R is Radius of earth (Take $\mathrm{R}=6400\,\mathrm{km}$). The depth d is equal to

Open solution

(A) 5260 km (B) 2560 km (C) 640 km (D) 4800 km

Q66. JEE Main 2023 (Online) 30th January Evening Shift

An object is allowed to fall from a height R above the earth, where R is the radius of earth. Its velocity when it strikes the earth's surface, ignoring air resistance, will be

Open solution

(A) $\sqrt{\frac{gR}{2}}$ (B) $\sqrt{gR}$ (C) $\sqrt{2gR}$ (D) $2\sqrt{gR}$

Q67. JEE Main 2023 (Online) 30th January Morning Shift

If the gravitational field in the space is given as $\left(-\frac{K}{r^2}\right)$. Taking the reference point to be at $\mathrm{r}=2\,\mathrm{cm}$ with gravitational potential $\mathrm{V}=10\,\mathrm{J/kg}$. Find the gravitational potential at $\mathrm{r}=3\,\mathrm{cm}$ in SI unit (Given, that $\mathrm{K}=6\,\mathrm{Jcm/kg}$)

Open solution

(A) 9 (B) 11 (C) 10 (D) 12

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Q68. JEE Main 2023 (Online) 29th January Evening Shift

The time period of a satellite of earth is 24 hours. If the separation between the earth and the satellite is decreased to one fourth of the previous value, then its new time period will become.

(A) 12 hours (B) 3 hours (C) 6 hours (D) 4 hours

Open solution

Q69. JEE Main 2023 (Online) 29th January Morning Shift

Two particles of equal mass 'm' move in a circle of radius 'r' under the action of their mutual gravitational attraction. The speed of each particle will be :

(A) $\sqrt{\frac{Gm}{4r}}$ (B) $\sqrt{\frac{Gm}{2r}}$ (C) $\sqrt{\frac{Gm}{r}}$ (D) $\sqrt{\frac{4Gm}{r}}$

Open solution

Q70. JEE Main 2023 (Online) 25th January Evening Shift

Every planet revolves around the sun in an elliptical orbit :-

A. The force acting on a planet is inversely proportional to square of distance from sun.

B. Force acting on planet is inversely proportional to product of the masses of the planet and the sun.

C. The Centripetal force acting on the planet is directed away from the sun.

D. The square of time period of revolution of planet around sun is directly proportional to cube of semi-major axis of elliptical orbit.

Choose the correct answer from the options given below :

(A) C and D only (B) B and C only (C) A and D only (D) A and C only

Open solution

Q71. JEE Main 2023 (Online) 25th January Evening Shift

A body of mass m is taken from earth surface to the height h equal to twice the radius of earth (R_e), the increase in potential energy will be :

(g = acceleration due to gravity on the surface of Earth)

(A) $\frac{1}{2}mgR_e$ (B) $3\,mgR_e$ (C) $\frac{1}{3}mgR_e$ (D) $\frac{2}{3}mgR_e$

Open solution

Q72. JEE Main 2023 (Online) 25th January Morning Shift

Assume that the earth is a solid sphere of uniform density and a tunnel is dug along its diameter throughout the earth. It is found that when a particle is released in this tunnel, it executes a simple harmonic motion. The mass of the particle is 100 g. The time period of the motion of the particle will be (approximately)

(Take g = 10 m s^{-2} , radius of earth = 6400 km)

(A) 12 hours (B) 1 hour 24 minutes

(C) 24 hours (D) 1 hour 40 minutes

Open solution

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Q73. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : A pendulum clock when taken to Mount Everest becomes fast.

Reason R : The value of g (acceleration due to gravity) is less at Mount Everest than its value on the surface of earth.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Both A and R are correct but R is NOT the correct explanation of A

(B) A is correct but R is not correct

(C) Both A and R are correct and R is the correct explanation of A

(D) A is not correct but R is correct

Q74. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

If the distance of the earth from Sun is 1.5×10^6 km. Then the distance of an imaginary planet from Sun, if its period of revolution is 2.83 years is :

(A) 6×10^6 km (B) 3×10^7 km (C) 6×10^7 km (D) 3×10^6 km

Q75. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

Given below are two statements:

Statement I : Acceleration due to earth's gravity decreases as you go 'up' or 'down' from earth's surface.

Statement II : Acceleration due to earth's gravity is same at a height 'h' and depth 'd' from earth's surface, if h = d.

In the light of above statements, choose the most appropriate answer from the options given below

(A) Statement I is correct but statement II is incorrect

(B) Both Statement I and II are correct

(C) Statement I is incorrect but statement II is correct

(D) Both Statement I and Statement II are incorrect

Q76. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

The weight of a body at the surface of earth is 18 N. The weight of the body at an altitude of 3200 km above the earth's surface is (given, radius of earth $R_e = 6400\ km$) :

(A) 9.8 N (B) 4.9 N (C) 19.6 N (D) 8 N

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Q77. JEE Main 2022 (Online) 30th July Evening Shift

The time period of X planet of our solar system is 8 years. The distance of earth from the sun is $1.5 \times 10^{11}\ \mathrm{m}$. The distance of the X planet from the sun will be :

(A) $3\sqrt{2} \times 10^{11}\ \mathrm{m}$ (B) $3 \times 10^{11}\ \mathrm{m}$

(C) $6 \times 10^{11}\ \mathrm{m}$ (D) $12 \times 10^{11}\ \mathrm{m}$

Open solution

Q78. JEE Main 2022 (Online) 29th July Evening Shift

An object of mass $1\ \mathrm{kg}$ is taken to a height from the surface of earth which is equal to three times the radius of earth. The gain in potential energy of the object will be [If, $\mathrm{g} = 10\ \mathrm{ms}^{-2}$ and radius of earth $= 6400\ \mathrm{km}$]

(A) 48 MJ (B) 24 MJ (C) 36 MJ (D) 12 MJ

Open solution

Q79. JEE Main 2022 (Online) 28th July Morning Shift

If the radius of earth shrinks by 2% while its mass remains same. The acceleration due to gravity on the earth's surface will approximately :

(A) decrease by 2% (B) decrease by 4% (C) increase by 2% (D) increase by 4%

Open solution

Q80. JEE Main 2022 (Online) 27th July Evening Shift

A body of mass m is projected with velocity $\lambda\, v_{\mathrm{e}}$ in vertically upward direction from the surface of the earth into space. It is given that v_{e} is escape velocity and $\lambda < 1$. If air resistance is considered to be negligible, then the maximum height from the centre of earth, to which the body can go, will be :

(R : radius of earth)

(A) $\frac{\mathrm{R}}{1+\lambda^2}$ (B) $\frac{R}{1-\lambda^2}$

(C) $\frac{R}{1-\lambda}$ (D) $\frac{\lambda^2 \mathrm{R}}{1-\lambda^2}$

Open solution

Q81. JEE Main 2022 (Online) 27th July Morning Shift

Two satellites A and B, having masses in the ratio $4 : 3$, are revolving in circular orbits of radii $3\mathrm{r}$ and $4\mathrm{r}$ respectively around the earth. The ratio of total mechanical energy of A to B is :

(A) 9 : 16 (B) 16 : 9 (C) 1 : 1 (D) 4 : 3

Open solution

Q82. JEE Main 2022 (Online) 26th July Evening Shift

A body is projected vertically upwards from the surface of earth with a velocity equal to one third of escape velocity. The maximum height attained by the body will be :

(Take radius of earth $= 6400\ \mathrm{km}$ and $\mathrm{g} = 10\ \mathrm{ms}^{-2}$)

(A) 800 km (B) 1600 km (C) 2133 km (D) 4800 km

Open solution

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Q83. JEE Main 2022 (Online) 26th July Morning Shift

The percentage decrease in the weight of a rocket, when taken to a height of $32\ \text{km}$ above the surface of earth will, be :

(Radius of earth $= 6400\ \text{km}$)

(A) 1% (B) 3% (C) 4% (D) 0.5%

Open solution

Q84. JEE Main 2022 (Online) 25th July Evening Shift

The length of a seconds pendulum at a height h = 2R from earth surface will be:

(Given R = Radius of earth and acceleration due to gravity at the surface of earth, g = π^2 ms^{-2})

(A) $\frac{2}{9}$ m (B) $\frac{4}{9}$ m (C) $\frac{8}{9}$ m (D) $\frac{1}{9}$ m

Open solution

Q85. JEE Main 2022 (Online) 25th July Evening Shift

An object is taken to a height above the surface of earth at a distance $\frac{5}{4}$ R from the centre of the earth. Where radius of earth, R = 6400 km. The percentage decrease in the weight of the object will be :

(A) 36% (B) 50% (C) 64% (D) 25%

Open solution

Q86. JEE Main 2022 (Online) 25th July Morning Shift

Three identical particles A, B and C of mass $100\ \text{kg}$ each are placed in a straight line with $\text{AB} = \text{BC} = 13\ \text{m}$. The gravitational force on a fourth particle P of the same mass is F, when placed at a distance $13\ \text{m}$ from the particle B on the perpendicular bisector of the line AC. The value of F will be approximately :

(A) 21 G (B) 100 G (C) 59 G (D) 42 G

Open solution

Q87. JEE Main 2022 (Online) 30th June Morning Shift

The radii of two planets A and B are in the ratio 2 : 3. Their densities are 3ρ and 5ρ respectively. The ratio of their acceleration due to gravity is :

(A) 9 : 4 (B) 9 : 8 (C) 9 : 10 (D) 2 : 5

Open solution

Q88. JEE Main 2022 (Online) 29th June Evening Shift

The time period of a satellite revolving around earth in a given orbit is 7 hours. If the radius of orbit is increased to three times its previous value, then approximate new time period of the satellite will be

(A) 40 hours (B) 36 hours (C) 30 hours (D) 25 hours

Open solution

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Q89. JEE Main 2022 (Online) 29th June Morning Shift

The escape velocity of a body on a planet 'A' is 12 kms^{-1}. The escape velocity of the body on another planet 'B', whose density is four times and radius is half of the planet 'A', is :

(A) 12 kms^{-1} (B) 24 kms^{-1} (C) 36 kms^{-1} (D) 6 kms^{-1}

Open solution

Q90. JEE Main 2022 (Online) 28th June Evening Shift

Two objects of equal masses placed at certain distance from each other attracts each other with a force of F. If one-third mass of one object is transferred to the other object, then the new force will be :

(A) $\frac{2}{9}$ F (B) $\frac{16}{9}$ F (C) $\frac{8}{9}$ F (D) F

Open solution

Q91. JEE Main 2022 (Online) 28th June Morning Shift

Two planets A and B of equal mass are having their period of revolutions T_A and T_B such that $T_A = 2T_B$. These planets are revolving in the circular orbits of radii r_A and r_B respectively. Which out of the following would be the correct relationship of their orbits?

(A) $2r_A^2 = r_B^3$ (B) $r_A^3 = 2r_B^3$

(C) $r_A^3 = 4r_B^3$ (D) $T_A^2 - T_B^2 = \frac{\pi^2}{GM}\left(r_B^3 - 4r_A^3\right)$

Open solution

Q92. JEE Main 2022 (Online) 27th June Evening Shift

The distance of the Sun from earth is 1.5×10^{11} m and its angular diameter is (2000)" when observed from the earth. The diameter of the Sun will be :

(A) 2.45×10^{10} m (B) 1.45×10^{10} m (C) 1.45×10^{9} m (D) 0.14×10^{9} m

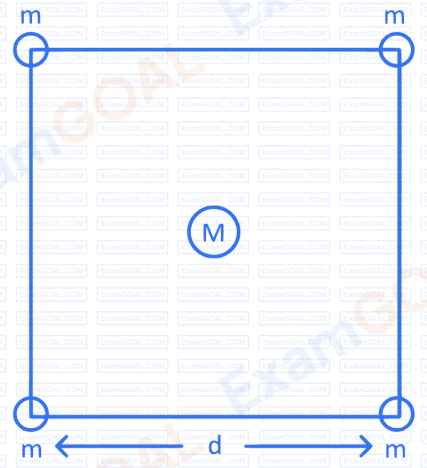
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Q93. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

Four spheres each of mass m from a square of side d (as shown in figure). A fifth sphere of mass M is situated at the centre of square. The total gravitational potential energy of the system is :



(A) $-\frac{Gm}{d}\left[(4+\sqrt{2})m+4\sqrt{2}M\right]$

(B) $-\frac{Gm}{d}\left[(4+\sqrt{2})M+4\sqrt{2}m\right]$

(C) $-\frac{Gm}{d}\left[3m^2+4\sqrt{2}M\right]$

(D) $-\frac{Gm}{d}\left[6m^2+4\sqrt{2}M\right]$

Q94. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

Given below are two statements :

Statement I : The law of gravitation holds good for any pair of bodies in the universe.

Statement II : The weight of any person becomes zero when the person is at the centre of the earth.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

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Q95. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : If we move from poles to equator, the direction of acceleration due to gravity of earth always points towards the center of earth without any variation in its magnitude.

Reason R : At equator, the direction of acceleration due to the gravity is towards the center of earth.

In the light of above statements, choose the correct answer from the options given below:

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

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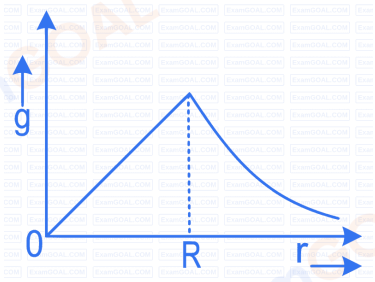
Q96. JEE Main 2022 (Online) 26th June Morning Shift

The variation of acceleration due to gravity (g) with distance (r) from the center of the earth is correctly represented by :

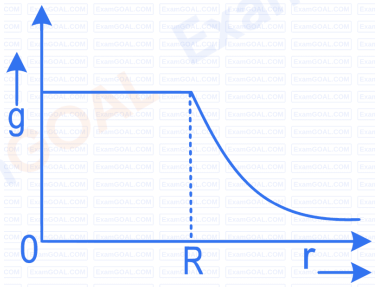
(Given R = radius of earth)

Open solution

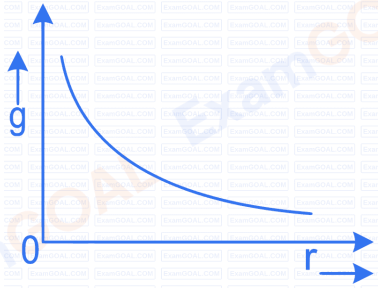
(A)



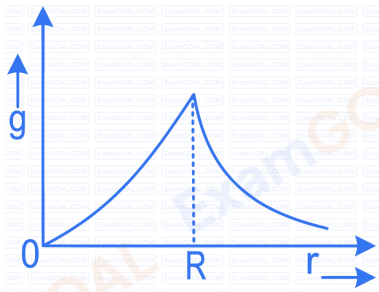
(B)



(C)



(D)



Q97. JEE Main 2022 (Online) 25th June Morning Shift

The height of any point P above the surface of earth is equal to diameter of earth. The value of acceleration due to gravity at point P will be : (Given g = acceleration due to gravity at the surface of earth).

Open solution

(A) g/2 (B) g/4 (C) g/3 (D) g/9

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Q98. JEE Main 2022 (Online) 24th June Evening Shift

The distance between Sun and Earth is R. The duration of year if the distance between Sun and Earth becomes 3R will be :

(A) $\sqrt{3}$ years (B) 3 years (C) 9 years (D) $3\sqrt{3}$ years

Open solution

Q99. JEE Main 2022 (Online) 24th June Morning Shift

The approximate height from the surface of earth at which the weight of the body becomes $\frac{1}{3}$ of its weight on the surface of earth is :

[Radius of earth R = 6400 km and $\sqrt{3}$ = 1.732]

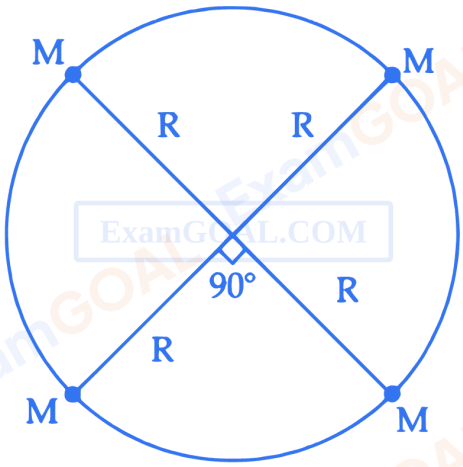
(A) 3840 km (B) 4685 km (C) 2133 km (D) 4267 km

Open solution

Q100. JEE Main 2021 (Online) 1st September Evening Shift

Four particles each of mass M, move along a circle of radius R under the action of their mutual gravitational attraction as shown in figure. The speed of each particle is :

Open solution



(A) $\frac{1}{2}\sqrt{\frac{GM}{R(2\sqrt{2}+1)}}$ (B) $\frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2}+1)}$

(C) $\frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2}-1)}$ (D) $\sqrt{\frac{GM}{R}}$

Q101. JEE Main 2021 (Online) 31st August Evening Shift

If R_E be the radius of Earth, then the ratio between the acceleration due to gravity at a depth 'r' below and a height 'r' above the earth surface is : (Given : r < R_E)

(A) $1-\frac{r}{R_E}-\frac{r^2}{R_E^2}-\frac{r^3}{R_E^3}$ (B) $1+\frac{r}{R_E}+\frac{r^2}{R_E^2}+\frac{r^3}{R_E^3}$

(C) $1+\frac{r}{R_E}-\frac{r^2}{R_E^2}+\frac{r^3}{R_E^3}$ (D) $1+\frac{r}{R_E}-\frac{r^2}{R_E^2}-\frac{r^3}{R_E^3}$

Open solution

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Q102. JEE Main 2021 (Online) 31st August Morning Shift

The masses and radii of the earth and moon are (M_1, R_1) and (M_2, R_2) respectively. Their centres are at a distance 'r' apart. Find the minimum escape velocity for a particle of mass 'm' to be projected from the middle of these two masses :

Open solution

(A) $V = \frac{1}{2}\sqrt{\frac{4G(M_1+M_2)}{r}}$

(B) $V = \sqrt{\frac{4G(M_1+M_2)}{r}}$

(C) $V = \frac{1}{2}\sqrt{\frac{2G(M_1+M_2)}{r}}$

(D) $V = \frac{\sqrt{2G}(M_1+M_2)}{r}$

Q103. JEE Main 2021 (Online) 27th August Evening Shift

A mass of 50 kg is placed at the centre of a uniform spherical shell of mass 100 kg and radius 50 m. If the gravitational potential at a point, 25 m from the centre is V kg/m. The value of V is :

Open solution

(A) -60 G

(B) +2 G

(C) -20 G

(D) -4 G

Q104. JEE Main 2021 (Online) 26th August Morning Shift

Inside a uniform spherical shell :

Open solution

(1) the gravitational field is zero

(2) the gravitational potential is zero

(3) the gravitational field is same everywhere

(4) the gravitational potential is same everywhere

(5) all of the above

Choose the most appropriate answer from the options given below :

(A) (1), (3) and (4) only

(B) (5) only

(C) (1), (2) and (3) only

(D) (2), (3) and (4) only

Q105. JEE Main 2021 (Online) 27th July Evening Shift

Two identical particles of mass 1 kg each go round a circle of radius R, under the action of their mutual gravitational attraction. The angular speed of each particle is :

Open solution

(A) $\sqrt{\frac{G}{2R^3}}$

(B) $\frac{1}{2}\sqrt{\frac{G}{R^3}}$

(C) $\frac{1}{2R}\sqrt{\frac{1}{G}}$

(D) $\frac{2G}{R^3}$

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Q106. JEE Main 2021 (Online) 27th July Evening Shift

The planet Mars has two moons, if one of them has a period 7 hours, 30 minutes and an orbital radius of 9.0×10^3 km. Find the mass of Mars.

$\left\{Given\ \frac{4\pi^2}{G} = 6 \times 10^{11} N^{-1} m^{-2} kg^2\right\}$

(A) 5.96×10^{19} kg (B) 3.25×10^{21} kg (C) 7.02×10^{25} kg (D) 6.00×10^{23} kg

Open solution

Q107. JEE Main 2021 (Online) 25th July Evening Shift

Consider a planet in some solar system which has a mass double the mass of earth and density equal to the average density of earth. If the weight of an object on earth is W, the weight of the same object on that planet will be :

(A) 2W (B) W (C) $2^{\frac{1}{3}}$W (D) $\sqrt{2}$W

Open solution

Q108. JEE Main 2021 (Online) 25th July Morning Shift

The minimum and maximum distances of a planet revolving around the sun are x_1 and x_2. If the minimum speed of the planet on its trajectory is v_0 then its maximum speed will be :

(A) $\frac{v_0 x_1^2}{x_2^2}$ (B) $\frac{v_0 x_2^2}{x_1^2}$ (C) $\frac{v_0 x_1}{x_2}$ (D) $\frac{v_0 x_2}{x_1}$

Open solution

Q109. JEE Main 2021 (Online) 22th July Evening Shift

A body is projected vertically upwards from the surface of earth with a velocity sufficient enough to carry it to infinity. The time taken by it to reach height h is _____ s.

(A) $\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{\frac{3}{2}}-1\right]$

(B) $\frac{1}{3}\sqrt{\frac{R_e}{2g}}\left[\left(1+\frac{h}{R_e}\right)^{\frac{3}{2}}-1\right]$

(C) $\sqrt{\frac{R_e}{2g}}\left[\left(1+\frac{h}{R_e}\right)^{\frac{3}{2}}-1\right]$

(D) $\frac{1}{3}\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{\frac{3}{2}}-1\right]$

Open solution

Q110. JEE Main 2021 (Online) 20th July Evening Shift

A satellite is launched into a circular orbit of radius R around earth, while a second satellite is launched into a circular orbit of radius 1.02 R. The percentage difference in the time periods of the two satellites is :

(A) 1.5 (B) 2.0 (C) 0.7 (D) 3.0

Open solution

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Q111. JEE Main 2021 (Online) 20th July Evening Shift

Consider a binary star system of star A and star B with masses m_A and m_B revolving in a circular orbit of radii r_A an r_B, respectively. If T_A and T_B are the time period of star A and star B, respectively,

Open solution

Then :

(A) $\frac{T_A}{T_B} = \left(\frac{r_A}{r_B}\right)^{\frac{3}{2}}$

(B) $T_A = T_B$

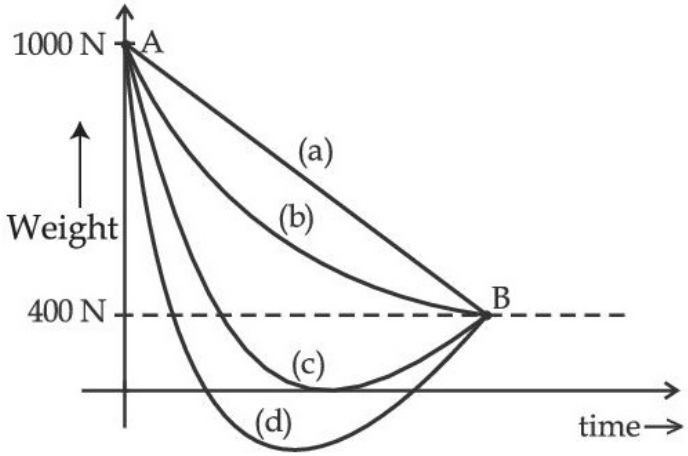
(C) $T_A > T_B$ (if $m_A > m_B$)

(D) $T_A > T_B$ (if $r_A > r_B$)

Q112. JEE Main 2021 (Online) 20th July Morning Shift

A person whose mass is 100 kg travels from Earth to Mars in a spaceship. Neglect all other objects in sky and take acceleration due to gravity on the surface of the Earth and Mars as 10 m/s^2 and 4 m/s^2 respectively. Identify from the below figures, the curve that fits best for the weight of the passenger as a function of time.

Open solution



(A) (b) (B) (c) (C) (d) (D) (a)

Q113. JEE Main 2021 (Online) 18th March Evening Shift

If the angular velocity of earth's spin is increased such that the bodies at the equator start floating, the duration of the day would be approximately : [Take g = 10 ms^{-2}, the radius of earth, R = 6400 $\times$ 10^3 m, Take π = 3.14]

Open solution

(A) 84 minutes (B) 1200 minutes (C) 60 minutes (D) does not change

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Q114. JEE Main 2021 (Online) 18th March Evening Shift

The angular momentum of a planet of mass M moving around the sun in an elliptical orbit is $\vec{L}$. The magnitude of the areal velocity of the planet is :

(A) $\frac{2L}{M}$ (B) $\frac{L}{2M}$ (C) $\frac{L}{M}$ (D) $\frac{4L}{M}$

Open solution

Q115. JEE Main 2021 (Online) 18th March Morning Shift

The time period of a satellite in a circular orbit of radius R is T. The period of another satellite in a circular orbit of radius 9R is :

(A) 9 T (B) 27 T (C) 12 T (D) 3 T

Open solution

Q116. JEE Main 2021 (Online) 17th March Evening Shift

A geostationary satellite is orbiting around an arbitrary planet 'P' at a height of 11R above the surface of 'P', R being the radius of 'P'. The time period of another satellite in hours at a height of 2R from the surface of 'P' is _____. 'P' has the time period of 24 hours.

(A) 3 (B) 5 (C) $6\sqrt{2}$ (D) $\frac{6}{\sqrt{2}}$

Open solution

Q117. JEE Main 2021 (Online) 16th March Morning Shift

The maximum and minimum distances of a comet from the Sun are 1.6×10^{12} m and 8.0×10^{10} m respectively. If the speed of the comet at the nearest point is 6×10^4 ms^{-1}, the speed at the farthest point is :

(A) 3.0×10^3 m/s (B) 6.0×10^3 m/s (C) 1.5×10^3 m/s (D) 4.5×10^3 m/s

Open solution

Q118. JEE Main 2021 (Online) 26th February Morning Shift

A planet revolving in elliptical orbit has :

A. a constant velocity of revolution.

B. has the least velocity when it is nearest to the sun.

C. its areal velocity is directly proportional to its velocity.

D. areal velocity is inversely proportional to its velocity.

E. to follow a trajectory such that the areal velocity is constant.

Choose the correct answer from the options given below :

(A) D only (B) E only (C) C only (D) A only

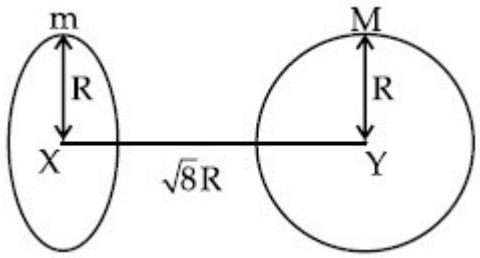
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Q119. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

Find the gravitational force of attraction between the ring and sphere as shown in the diagram, where the plane of the ring is perpendicular to the line joining the centres. If $\sqrt{8}$R is the distance between the centres of a ring (of mass 'm') and a sphere (mass 'M') where both have equal radius 'R'.



(A) $\frac{2\sqrt{2}}{3}.\frac{GMm}{R^2}$

(B) $\frac{\sqrt{8}}{9}.\frac{GmM}{R}$

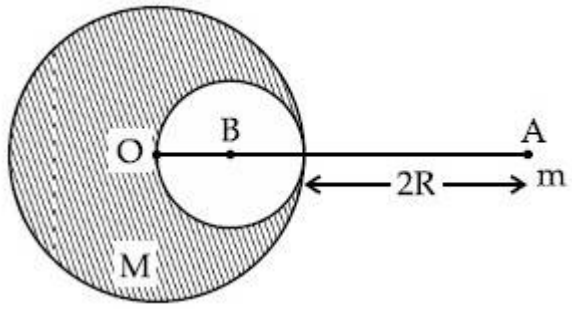
(C) $\frac{\sqrt{8}}{27}.\frac{GmM}{R^2}$

(D) $\frac{1}{3\sqrt{8}}.\frac{GMm}{R^2}$

Q120. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

A solid sphere of radius R gravitationally attracts a particle placed at 3R from its centre with a force F_1. Now a spherical cavity of radius $\left(\frac{R}{2}\right)$ is made in the sphere (as shown in figure) and the force becomes F_2. The value of $F_1 : F_2$ is



(A) 36 : 25

(B) 41 : 50

(C) 50 : 41

(D) 25 : 36

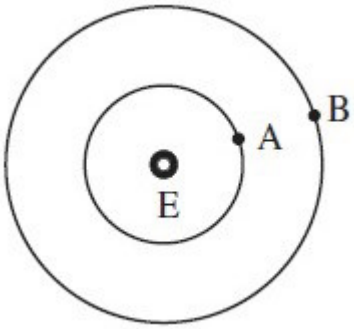
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Q121. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

Two satellites A and B of masses 200 kg and 400 kg are revolving round the earth at height of 600 km and 1600 km respectively.

If T_A and T_B are the time periods of A and B respectively then the value of $T_B - T_A$:



[Given : radius of earth = 6400 km, mass of earth = 6×10^{24} kg]

(A) 1.33×10^3 s (B) 4.24×10^2 s (C) 3.33×10^2 s (D) 4.24×10^3 s

Q122. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : The escape velocities of planet A and B are same. But A and B are of unequal mass.

Reason R : The product of their mass and radius must be same. $M_1R_1 = M_2R_2$

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are correct and R is the correct explanation of A

(B) Both A and R are correct but R is NOT the correct explanation of A

(C) A is correct but R is not correct

(D) A is not correct but R is correct

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Q123. JEE Main 2021 (Online) 24th February Evening Shift

A body weights 49N on a spring balance at the north pole. What will be its weight recorded on the same weighing machine, if it is shifted to the equator?

[Use $g = \frac{GM}{R^2}$ = 9.8 ms^{-2} and radius of earth, R = 6400 km.]

(A) 49 N (B) 49.83 N (C) 48.83 N (D) 49.17 N

Open solution

Q124. JEE Main 2021 (Online) 24th February Morning Shift

Four identical particles of equal masses 1 kg made to move along the circumference of a circle of radius 1 m under the action of their own mutual gravitational attraction. The speed of each particle will be :

(A) $\sqrt{\frac{G}{2}(1+2\sqrt{2})}$ (B) $\sqrt{\frac{G}{2}(2\sqrt{2}-1)}$

(C) $\sqrt{G(1+2\sqrt{2})}$ (D) $\frac{1}{2}\sqrt{G(1+2\sqrt{2})}$

Open solution

Q125. JEE Main 2021 (Online) 24th February Morning Shift

Consider two satellites S_1 and S_2 with periods of revolution 1 hr. and 8 hr. respectively revolving around a planet in circular orbits. The ratio of angular velocity of satellite S_1 to the angular velocity of satellite S_2 is :

(A) 1 : 4 (B) 8 : 1 (C) 2 : 1 (D) 1 : 8

Open solution

Q126. JEE Main 2021 (Online) 24th February Morning Shift

Two stars of masses m and 2m at a distance d rotate about their common centre of mass in free space. The period of revolution is :

(A) $\frac{1}{2\pi}\sqrt{\frac{d^3}{3Gm}}$ (B) $2\pi\sqrt{\frac{3Gm}{d^3}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{3Gm}{d^3}}$ (D) $2\pi\sqrt{\frac{d^3}{3Gm}}$

Open solution

Q127. JEE Main 2020 (Online) 6th September Evening Slot

Two planets have masses M and 16 M and their radii are a and $2a$, respectively. The separation between the centres of the planets is $10a$. A body of mass m is fired from the surface of the larger planet towards the smaller planet along the line joining their centres. For the body to be able to reach at the surface of smaller planet, the minimum firing speed needed is :

(A) $2\sqrt{\frac{GM}{a}}$ (B) $\sqrt{\frac{GM^2}{ma}}$

(C) $\frac{3}{2}\sqrt{\frac{5GM}{a}}$ (D) $4\sqrt{\frac{GM}{a}}$

Open solution

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Q128. JEE Main 2020 (Online) 6th September Morning Slot

A satellite is in an elliptical orbit around a planet P. It is observed that the velocity of the satellite when it is farthest from the planet is 6 times less than that when it is closest to the planet. The ratio of distances between the satellite and the planet at closest and farthest points is:

Open solution

(A) 1 : 2 (B) 1 : 3 (C) 1 : 6 (D) 3 : 4

Q129. JEE Main 2020 (Online) 5th September Evening Slot

The acceleration due to gravity on the earth's surface at the poles is g and angular velocity of the earth about the axis passing through the pole is ω. An object is weighed at the equator and at a height h above the poles by using a spring balance. If the weights are found to be same, then h is (h << R, where R is the radius of the earth)

Open solution

(A) $\frac{R^2\omega^2}{2g}$ (B) $\frac{R^2\omega^2}{g}$ (C) $\frac{R^2\omega^2}{8g}$ (D) $\frac{R^2\omega^2}{4g}$

Q130. JEE Main 2020 (Online) 5th September Morning Slot

The value of the acceleration due to gravity is g_1 at a height h = $\frac{R}{2}$ (R = radius of the earth) from the surface of the earth. It is again equal to g_1 at a depth d below the surface of the earth. The ratio $\left(\frac{d}{R}\right)$ equals :

Open solution

(A) $\frac{5}{9}$ (B) $\frac{1}{9}$ (C) $\frac{7}{9}$ (D) $\frac{4}{9}$

Q131. JEE Main 2020 (Online) 4th September Evening Slot

A body is moving in a low circular orbit about a planet of mass M and radius R. The radius of the orbit can be taken to be R itself. Then the ratio of the speed of this body in the orbit to the escape velocity from the planet is:

Open solution

(A) 2 (B) 1 (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

Q132. JEE Main 2020 (Online) 4th September Morning Slot

On the x-axis and at a distance x from the origin, the gravitational field due a mass distribution is given by $\frac{Ax}{(x^2+a^2)^{3/2}}$ in the x-direction. The magnitude of gravitational potential on the x-axis at a distance x, taking its value to be zero at infinity, is:

Open solution

(A) $A\left(x^2+a^2\right)^{3/2}$ (B) $A\left(x^2+a^2\right)^{1/2}$

(C) $\frac{A}{(x^2+a^2)^{1/2}}$ (D) $\frac{A}{(x^2+a^2)^{3/2}}$

Q133. JEE Main 2020 (Online) 3rd September Evening Slot

The mass density of a planet of radius R varies with the distance r from its centre as $\rho(r) = \rho_0\left(1-\frac{r^2}{R^2}\right)$.

Then the gravitational field is maximum at :

Open solution

(A) $r=\frac{1}{\sqrt{3}}R$ (B) r = R (C) $r=\sqrt{\frac{3}{4}}R$ (D) $r=\sqrt{\frac{5}{9}}R$

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Q134. JEE Main 2020 (Online) 3rd September Morning Slot

A satellite is moving in a low nearly circular orbit around the earth. Its radius is roughly equal to that of the earth's radius R_e . By firing rockets attached to it, its speed is instantaneously increased in the direction of its motion so that it become $\sqrt{\frac{3}{2}}$ times larger. Due to this the farthest distance from the centre of the earth that the satellite reaches is R. Value of R is :

(A) $2R_e$ (B) $3R_e$ (C) $4R_e$ (D) $2.5R_e$

Open solution

Q135. JEE Main 2020 (Online) 2nd September Evening Slot

The height 'h' at which the weight of a body will be the same as that at the same depth 'h' from the surface of the earth is (Radius of the earth is R and effect of the rotation of the earth is neglected)

(A) $\frac{R}{2}$ (B) $\frac{\sqrt{5}R-R}{2}$ (C) $\frac{\sqrt{3}R-R}{2}$ (D) $\frac{\sqrt{5}}{2}R-R$

Open solution

Q136. JEE Main 2020 (Online) 2nd September Morning Slot

The mass density of a spherical galaxy varies as $\frac{K}{r}$ over a large distance 'r' from its centre. In that region, a small star is in a circular orbit of radius R. Then the period of revolution, T depends on R as :

(A) $T^2 \propto R$ (B) $T^2 \propto R^3$

(C) $T \propto R$ (D) $T^2 \propto \frac{1}{R^3}$

Open solution

Q137. JEE Main 2020 (Online) 9th January Evening Slot

Planet A has mass M and radius R. Planet B has half the mass and half the radius of Planet A. If the escape velocities from the Planets A and B are v_A and v_B, respectively, then $\frac{v_A}{v_B} = \frac{n}{4}$. The value of n is :

(A) 1 (B) 2 (C) 4 (D) 3

Open solution

Q138. JEE Main 2020 (Online) 9th January Morning Slot

A body A of mass m is moving in a circular orbit of radius R about a planet. Another body B of mass $\frac{m}{2}$ collides with A with a velocity which is half $\left(\frac{\vec{v}}{2}\right)$ the instantaneous velocity $\vec{v}$ of A. The collision is completely inelastic. Then, the combined body :

(A) starts moving in an elliptical orbit around the planet.

(B) Falls vertically downwards towards the planet

(C) Escapes from the Planet's Gravitational field.

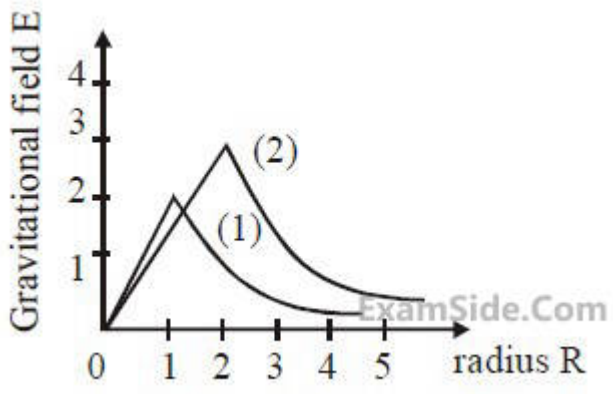
(D) continues to move in a circular orbit

Open solution

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Q139. JEE Main 2020 (Online) 8th January Morning Slot

Consider two solid spheres of radii R_1 = 1m, R_2 = 2m and masses M_1 and M_2, respectively. The gravitational field due to sphere (1) and (2) are shown. The value of $\frac{M_1}{M_2}$ is :



(A) $\frac{2}{3}$ (B) $\frac{1}{6}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

Open solution

Q140. JEE Main 2020 (Online) 7th January Evening Slot

A box weight 196 N on a spring balance at the north pole. Its weight recorded on the same balance if it is shifted to the equator is close to (Take g = 10 ms^{-2} at the north pole and the radius of the earth = 6400 km) :

(A) 194.32 N (B) 195.66 N (C) 195.32 N (D) 194.66 N

Open solution

Q141. JEE Main 2020 (Online) 7th January Morning Slot

A satellite of mass m is launched vertically upwards with an initial speed u from the surface of the earth. After it reaches height R (R = radius of the earth), it ejects a rocket of mass $\frac{m}{10}$ so that subsequently the satellite moves in a circular orbit. The kinetic energy of the rocket is (G is the gravitational constant; M is the mass of the earth) :

(A) $\frac{3m}{8}\left(u+\sqrt{\frac{5GM}{6R}}\right)^2$ (B) $\frac{m}{20}\left(u^2+\frac{113}{100}\frac{GM}{R}\right)$

(C) $5m\left(u^2-\frac{119}{200}\frac{GM}{R}\right)$ (D) $\frac{m}{20}\left(u-\sqrt{\frac{2GM}{3R}}\right)^2$

Open solution

Q142. JEE Main 2019 (Online) 12th April Evening Slot

The ratio of the weights of a body on the Earth's surface to that on the surface of a planets is 9 : 4. The mass of the planet is $\frac{1}{9}$ th of that of the Earth. If 'R' is the radius of the Earth, what is the radius of the planet ? (Take the planets to have the same mass density)

(A) $\frac{R}{9}$ (B) $\frac{R}{2}$ (C) $\frac{R}{3}$ (D) $\frac{R}{4}$

Open solution

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Q143. JEE Main 2019 (Online) 10th April Evening Slot

A spaceship orbits around a planet at a height of 20 km from its surface. Assuming that only gravitational field of the planet acts on the spaceship, what will be the number of complete revolutions made by the spaceship in 24 hours around the planet?

[Given ; Mass of planet = 8×10^{22} kg, Radius of planet = 2×10^{6} m, Gravitational constant G = 6.67×10^{-11} Nm^2/kg^2]

(A) 13 (B) 9 (C) 17 (D) 11

Open solution

Q144. JEE Main 2019 (Online) 10th April Morning Slot

The value of acceleration due to gravity at Earth's surface is 9.8 ms^{-2}. The altitude above its surface at which the acceleration due to gravity decreases to 4.9 ms^{-2}, is close to : (Radius of earth = 6.4×10^{6} m)

(A) 1.6×10^{6} m (B) 9.0×10^{6} m (C) 6.4×10^{6} m (D) 2.6×10^{6} m

Open solution

Q145. JEE Main 2019 (Online) 9th April Evening Slot

A test particle is moving in a circular orbit in the gravitational field produced by a mass density $\rho(r) = \frac{K}{r^2}$. Identify the correct relation between the radius R of the particle's orbit and its period T

(A) T^2/R^3 is a constant (B) TR is a constant

(C) T/R^2 is a constant (D) T/R is a constant

Open solution

Q146. JEE Main 2019 (Online) 9th April Morning Slot

A solid sphere of mass 'M' and radius 'a' is surrounded by a uniform concentric spherical shell of thickness 2a and mass 2M. The gravitational field at distance '3a' from the centre will be :

(A) $\frac{GM}{3a^2}$ (B) $\frac{2GM}{9a^2}$ (C) $\frac{GM}{9a^2}$ (D) $\frac{2GM}{3a^2}$

Open solution

Q147. JEE Main 2019 (Online) 8th April Evening Slot

A rocket has to be launched from earth in such a way that it never returns. If E is the minimum energy delivered by the rocket launcher, what should be the minimum energy that the launcher should have if the same rocket is to be launched from the surface of the moon ? Assume that the density of the earth and the moon are equal and that the earth's volume is 64 times the volume of the moon :-

(A) E/32 (B) E/16 (C) E/4 (D) E/64

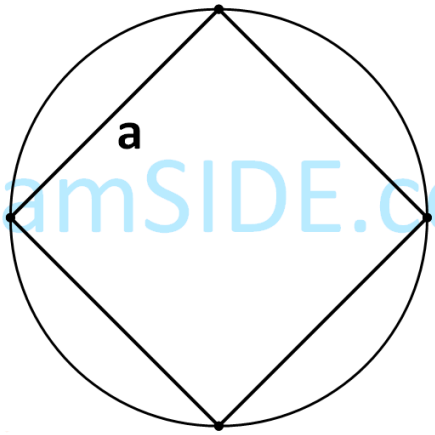
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Q148. JEE Main 2019 (Online) 8th April Morning Slot

Four identical particles of mass M are located at the corners of a square of side 'a'. What should be their speed if each of them revolves under the influence of other's gravitational field in a circular orbit circumscribing the square?

Open solution



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(A) $1.21\sqrt{\frac{GM}{a}}$ (B) $1.16\sqrt{\frac{GM}{a}}$ (C) $1.41\sqrt{\frac{GM}{a}}$ (D) $1.35\sqrt{\frac{GM}{a}}$

Q149. JEE Main 2019 (Online) 12th January Evening Slot

Two satellites, A and B, have masses m and 2m respectively. A is in a circular orbit of radius R, and B is in a circular orbit of radius 2R around the earth. The ratio of their kinetic energies, T_A/T_B, is ;

Open solution

(A) 2 (B) $\frac{1}{2}$ (C) $\sqrt{\frac{1}{2}}$ (D) 1

Q150. JEE Main 2019 (Online) 12th January Morning Slot

A satellite of mass M is in a circular orbit of radius R about the centre of the earth. A meteorite of the same mass, falling towards the earth, collides with the satellite completely inelastically. The speeds of the satellite and the meteorite are the same, just before the collision. The subsequent motion of the combined body will be :

Open solution

(A) in the same circular orbit of radius R

(B) such that it escapes to infinity

(C) in a circular orbit of a different radius

(D) in an elliptical orbit

Q151. JEE Main 2019 (Online) 12th January Morning Slot

A straight rod of length L extends from x = a to x = L + a. The gravitational force it exerts on a point mass 'm' at x = 0, if the mass per unit length of the rod is A + Bx^2 , is given by :

Open solution

(A) $Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)-BL\right]$ (B) $Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)+BL\right]$

(C) $Gm\left[A\left(\frac{1}{a+L}-\frac{1}{a}\right)+BL\right]$ (D) $Gm\left[A\left(\frac{1}{a+L}-\frac{1}{a}\right)-BL\right]$

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Q152. JEE Main 2019 (Online) 11th January Morning Slot

A satellite is revolving in a circular orbit at a height h from the earth's surface, such that h < < R where R is the radius of the earth. Assuming that the effect of earth's atmosphere can be neglected the minimum increase in the speed required so that the satellite could escape from the gravitational field of earth is :

Open solution

(A) $\sqrt{gR}\left(\sqrt{2}-1\right)$ (B) $\sqrt{2gR}$

(C) $\sqrt{gR}$ (D) $\frac{\sqrt{gR}}{2}$

Q153. JEE Main 2019 (Online) 10th January Evening Slot

Two stars of masses 3×10^{31} kg each, and at distance 2×10^{11} m rotate in a plane about their common centre of mass O. A meteorite passes through O moving perpendicular to the star's rotation plane. In order to escape from the gravitational field of this double star, the minimum speed that meteorite should have at O is - (Take Gravitational constant; G = 6.67×10^{-11} $Nm^2 kg^{-2}$)

Open solution

(A) 2.4×10^4 m/s (B) 1.4×10^5 m/s (C) 3.8×10^4 m/s (D) 2.8×10^5 m/s

Q154. JEE Main 2019 (Online) 10th January Morning Slot

A satellite is moving with a constant speed v in circular orbit around the earth. An object of mass 'm' is ejected from the satellite such that it just escapes from the gravitational pull of the earth. At the time of ejection, the kinetic energy of the object is -

Open solution

(A) mv^2 (B) $\frac{1}{2}mv^2$ (C) $\frac{3}{2}mv^2$ (D) $2mv^2$

Q155. JEE Main 2019 (Online) 9th January Evening Slot

The energy required to take a satellite to a height 'h' above Earth surface (radius of Earth = 6.4×10^3 km) is E_1 and kinetic energy required for the satellite to be in a circular orbit at this height is E_2. The value of h for which E_1 and E_2 are equal, is

Open solution

(A) 1.6×10^3 km (B) 3.2×10^3 km (C) 6.4×10^3 km (D) 1.28×10^4 km

Q156. JEE Main 2019 (Online) 9th January Morning Slot

If the angular momentum of a planet of mass m, moving around the Sun in a circular orbit is L, about the center of the Sun, its areal velocity is :

Open solution

(A) $\frac{L}{m}$ (B) $\frac{4L}{m}$ (C) $\frac{L}{2m}$ (D) $\frac{2L}{m}$

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Q157. JEE Main 2018 (Online) 16th April Morning Slot

Open solution

Suppose that the angular velocity of rotation of earth is increased. Then, as a consequence :

(A) Weight of the object, everywhere on the earth, will increase.

(B) Weight of the object, everywhere on the earth, will decrease.

(C) There will be no change in weight anywhere on the earth.

(D) Except at poles, weight of the object on the earth will decrease.

Q158. JEE Main 2018 (Offline)

Open solution

A particle is moving with a uniform speed in a circular orbit of radius R in a central force inversely proportional to the nth power of R. If the period of rotation of the particle is T, then :

(A) $T \propto R^{n/2}$

(B) $T \propto R^{3/2}$ for any n

(C) $T \propto R^{n/2+1}$

(D) $T \propto R^{(n+1)/2}$

Q159. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

A body of mass m is moving in a circular orbit of radius R about a planet of mass M. At some instant, it splits into two equal masses. The first mass moves in a circular orbit of radius $\frac{R}{2}$, and the other mass, in a circular orbit of radius $\frac{3R}{2}$. The difference between the final and initial total energies is :

(A) $-\frac{GMm}{2R}$

(B) $+\frac{GMm}{6R}$

(C) $\frac{GMm}{2R}$

(D) $-\frac{GMm}{6R}$

Q160. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

Take the mean distance of the moon and the sun from the earth to be 0.4×10^6 km and 150×10^6 km respectively. Their masses are 8×10^{22} kg and 2×10^{30} kg respectively. The radius of the earth is 6400 km. Let ΔF_1 be the difference in the forces exerted by the moon at the nearest and farthest points on the earth and ΔF_2 be the difference in the force exerted by the sun at the nearest and farthest points on the earth. Then, the number closest to $\frac{\Delta F_1}{\Delta F_2}$ is :

(A) 2

(B) 10^{-2}

(C) 0.6

(D) 6

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Q161. JEE Main 2017 (Online) 9th April Morning Slot

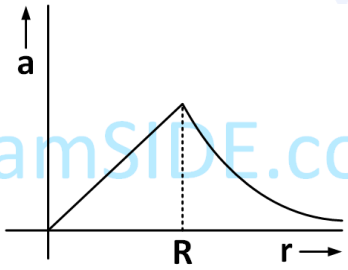
Open solution

The mass density of a spherical body is given by
$\rho(r) = \frac{k}{r}$ for $r \leq R$ and $\rho(r) = 0$ for $r > R$,

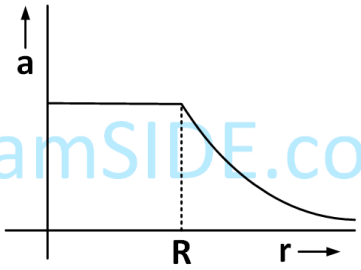
where r is the distance from the centre.

The correct graph that describes qualitatively the acceleration, a, of a test particle as a function of r is :

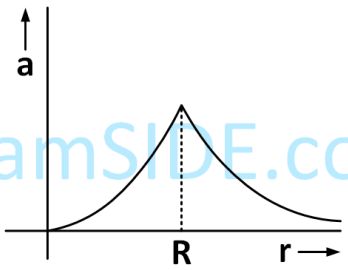
(A)



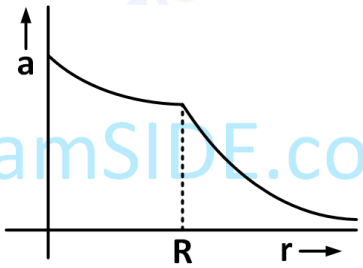
(B)



(C)



(D)



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Q162. JEE Main 2017 (Online) 8th April Morning Slot

Open solution

If the Earth has no rotational motion, the weight of a person on the equator is W. Determine the speed with which the earth would have to rotate about its axis so that the person at the equator will weigh $\frac{3}{4}$ W. Radius of the Earth is 6400 km and g=10 m/s^2.

(A) 1.1×10^{-3} rad/s

(B) 0.83×10^{-3} rad/s

(C) 0.63×10^{-3} rad/s

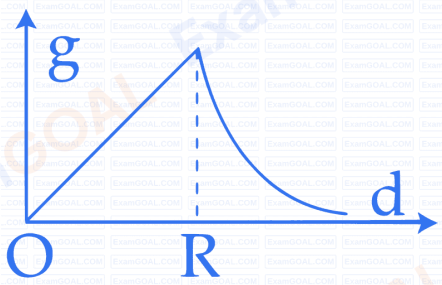
(D) 0.28×10^{-3} rad/s

Q163. JEE Main 2017 (Offline)

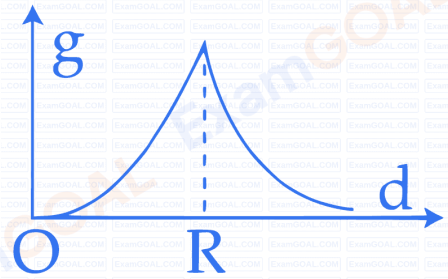
Open solution

The variation of acceleration due to gravity g with distance d from centre of the earth is best represented by (R = Earth's radius):

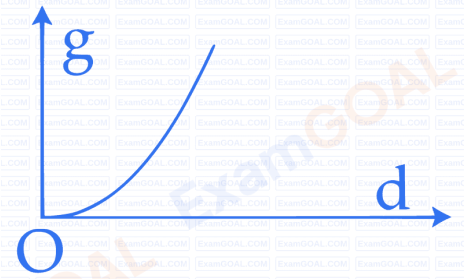
(A)



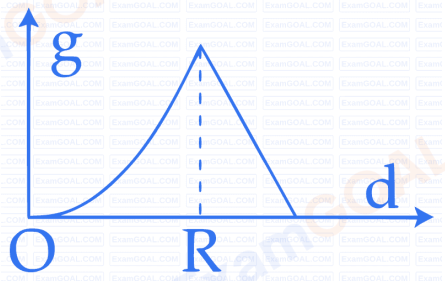
(B)



(C)



(D)

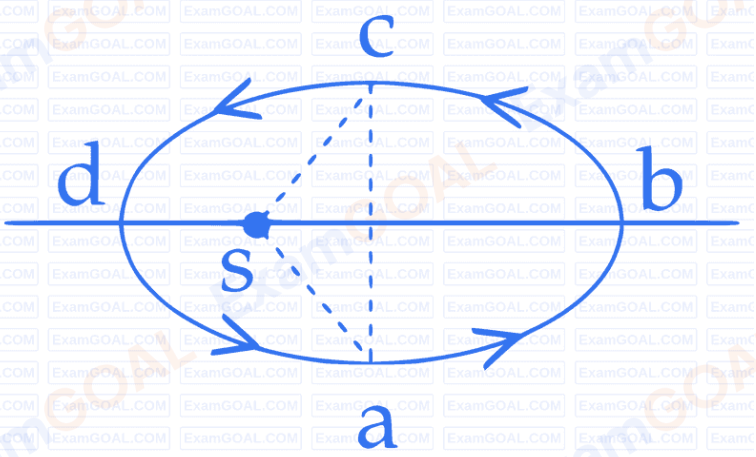


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Q164. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

Figure shows elliptical path abcd of a planet around the sun S such that the area of triangle csa is $\frac{1}{4}$ the area of the ellipse. (See figure) With db as the semimajor axis, and ca as the semiminor axis. If t_1 is the time taken for planet to go over path abc and t_2 for path taken over cda then :



(A) $t_1 = t_2$ (B) $t_1 = 2t_2$ (C) $t_1 = 3t_2$ (D) $t_1 = 4t_2$

Q165. JEE Main 2016 (Offline)

Open solution

A satellite is revolving in a circular orbit at a height $'h'$ from the earth's surface (radius of earth $R; h << R$). The minimum increase in its orbital velocity required, so that the satellite could escape from the earth's gravitational field, is close to : (Neglect the effect of atmosphere.)

(A) $\sqrt{2gR}$ (B) $\sqrt{gR}$ (C) $\sqrt{gR/2}$ (D) $\sqrt{gR}(\sqrt{2}-1)$

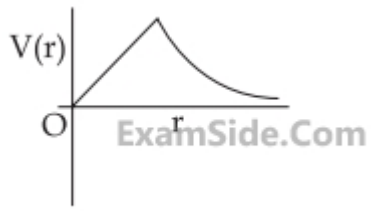
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Q166. JEE Main 2015 (Online) 11th April Morning Slot

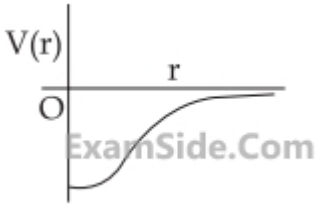
Which of the following most closely depicts the correct variation of the gravitation potential V(r) due to a large planet of radius R and uniform mass density ? *(figures are not drawn to scale)*

Open solution

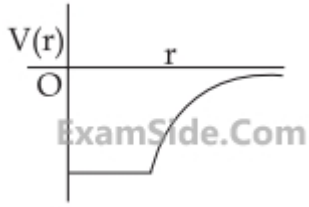
(A)



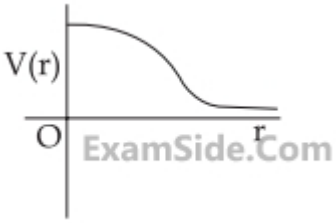
(B)



(C)



(D)



Q167. JEE Main 2015 (Online) 10th April Morning Slot

A very long (length L) cylindrical galaxy is made of uniformly distributed mass and has radius R (R << L). A star outside the galaxy is orbiting the galaxy in a plane perpendicular to the galaxy and passing through its centre. If the time period of star is T and its distance from the galaxy's axis is r, then :

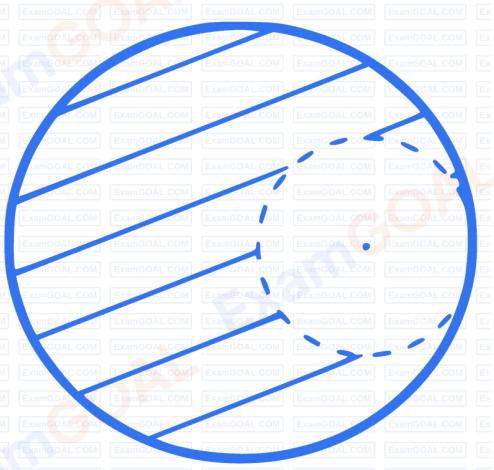
Open solution

(A) $T^2 \propto r^3$ (B) $T \propto r^2$ (C) $T \propto r$ (D) $T \propto \sqrt{r}$

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Q168. JEE Main 2015 (Offline)

From a solid sphere of mass M and radius R, a spherical portion of radius $R/2$ is removed, as shown in the figure. Taking gravitational potential $V = 0$ at $r = \infty$, the potential at the center of the cavity thus formed is:
$(G = gravitational\ constant)$



(A) $\frac{-2GM}{3R}$ (B) $\frac{-2GM}{R}$ (C) $\frac{-GM}{2R}$ (D) $\frac{-GM}{R}$

Open solution

Q169. JEE Main 2014 (Online) 19th April Morning Slot

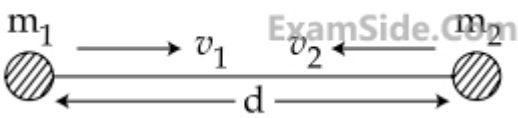
The gravitational field in a region is given by $\vec{g} = 5N/kg\hat{i} + 12N/kg\hat{j}$. The change in the gravitational potential energy of a particle of mass 1 kg when it is taken from the origin to a point (7m, -3m) is :

(A) 71 J (B) 13 $\sqrt{58}$ J (C) $-$ 71 J (D) 1 J

Open solution

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Q170. JEE Main 2014 (Online) 12th April Morning Slot



Open solution

Two hypothetical planets of masses m_1 and m_2 are at rest when they are infinite distance apart. Because of the gravitational force they move towards each other along the line joining their centres. What is their speed when their separation is 'd' ?
(Speed of m_1 is v_1 and that of m_2 is v_2)

(A) $v_1 = v_2$

(B) $v_1 = m_2\sqrt{\frac{2G}{d(m_1+m_2)}}$

$v_2 = m_1\sqrt{\frac{2G}{d(m_1+m_2)}}$

(C) $v_1 = m_1\sqrt{\frac{2G}{d(m_1+m_2)}}$

$v_2 = m_2\sqrt{\frac{2G}{d(m_1+m_2)}}$

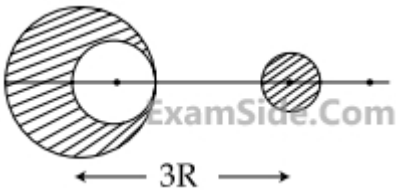
(D) $v_1 = m_2\sqrt{\frac{2G}{m_1}}$

$v_2 = m_1\sqrt{\frac{2G}{m_2}}$

Q171. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

From a sphere of mass M and radius R, a smaller sphere of radius $\frac{R}{2}$ is carved out such that the cavity made in the original sphere is between its centre and the periphery. (See figure). For the configuration in the figure where the distance between the center of the original sphere and the removed sphere is 3R, the gravitational force between the two spheres is :



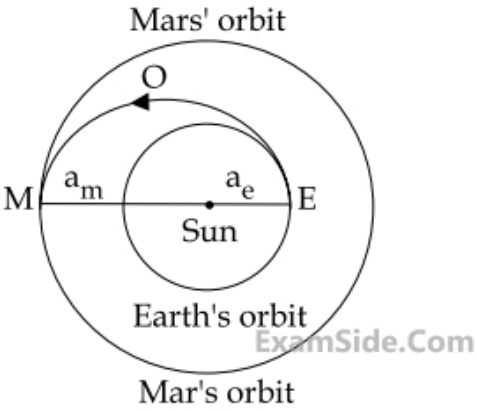
(A) $\frac{41\,GM^2}{3600R^2}$ (B) $\frac{41\,GM^2}{450R^2}$ (C) $\frac{59\,GM^2}{450R^2}$ (D) $\frac{GM^2}{225R^2}$

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Q172. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

India's Mangalyan was sent to the Mars by launching it into a transfer orbit EOM around the sun. It leaves the earth at E and meets Mars at M. If the semi-major axis of earth's orbit is $a_e = 1.5 \times 10^{11}$ m, that of Mars' orbit $a_m = 2.28 \times 10^{11}$ m, then Kepler's laws give the estimate of time for Mangalyan to reach Mars from Earth to be close to :



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(A) 500 days (B) 320 days (C) 260 days (D) 220 days

Q173. JEE Main 2014 (Offline)

Open solution

Four particles, each of mass M and equidistant from each other, move along a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is :

(A) $\sqrt{\frac{GM}{R}}$ (B) $\sqrt{2\sqrt{2}\frac{GM}{R}}$

(C) $\sqrt{\frac{GM}{R}\left(1+2\sqrt{2}\right)}$ (D) $\frac{1}{2}\sqrt{\frac{GM}{R}\left(1+2\sqrt{2}\right)}$

Q174. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

The gravitational field in a region is given by: $\vec{E} = (5N/kg)\hat{i} + (12N/kg)\hat{j}$. If the potential at the origin is taken to be zero, then the ratio of the potential at the points (12 m, 0) and (0, 5 m) is :

(A) Zero (B) 1 (C) $\frac{144}{25}$ (D) $\frac{25}{144}$

Q175. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

The change in the value of acceleration of earth towards sun, when the moon comes from the position of solar eclipse to the position on the other side of earth in line with sun is: (mass of the moon = 7.36×10^{22} kg, radius of the moon's orbit = 3.8×10^8 m).

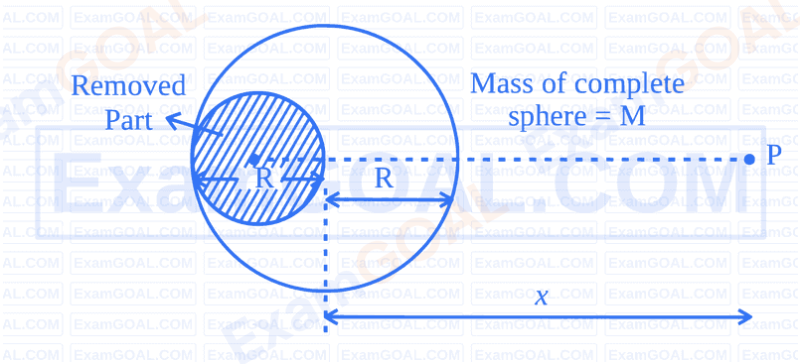
(A) 6.73×10^{-5} m/s^2 (B) 6.73×10^{-3} m/s^2 (C) 6.73×10^{-2} m/s^2 (D) 6.73×10^{-4} m/s^2

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Q176. JEE Main 2013 (Online) 9th April Morning Slot

The gravitational field, due to the 'left over part' of a uniform sphere (from which a part as shown, has been 'removed out'), at a very far off point, P, located as shown, would be (nearly) :

Open solution



(A) $\frac{5}{6}\frac{GM}{x^2}$ (B) $\frac{8}{9}\frac{GM}{x^2}$ (C) $\frac{7}{8}\frac{GM}{x^2}$ (D) $\frac{6}{7}\frac{GM}{x^2}$

Q177. JEE Main 2013 (Offline)

What is the minimum energy required to launch a satellite of mass m from the surface of a planet of mass M and radius R in a circular orbit at an altitude of $2R$?

Open solution

(A) $\frac{5GmM}{6R}$ (B) $\frac{2GmM}{3R}$ (C) $\frac{GmM}{2R}$ (D) $\frac{GmM}{3R}$

Q178. AIEEE 2012 (Online) 26th May

A point particle is held on the axis of a ring of mass m and radius r at a distance r from its centre C. When released, it reaches C under the gravitational attraction of the ring. Its speed at C will be

Open solution

(A) $\sqrt{\frac{2Gm}{r}\left(\sqrt{2}-1\right)}$ (B) $\sqrt{\frac{Gm}{r}}$

(C) $\sqrt{\frac{2Gm}{r}\left(1-\frac{1}{\sqrt{2}}\right)}$ (D) $\sqrt{\frac{2Gm}{r}}$

Q179. AIEEE 2012 (Online) 19th May

Two point masses of mass $m_1 = fM$ and $m_2 = (1-f)M$ $(f<1)$ are in outer space (far from gravitational influence of other objects) at a distance R from each other. They move in circular orbits about their centre of mass with angular velocities ω_1 for m_1 and ω_2 for m_2. In that case

Open solution

(A) $(1-f)\omega_1 = f\omega_2$ (B) $\omega_1 = \omega_2$ and independent of f

(C) $f\omega_1 = (1-f)\omega_2$ (D) $\omega_1 = \omega_2$ and depend on f

Q180. AIEEE 2012 (Online) 12th May

Assuming the earth to be a sphere of uniform density, the acceleration due to gravity inside the earth at a distance of r from the centre is proportional to

Open solution

(A) r (B) r^{-1} (C) r^2 (D) r^{-2}

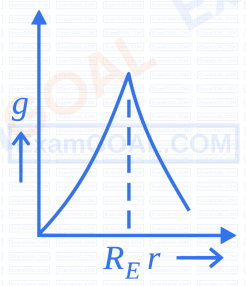
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Q181. AIEEE 2012 (Online) 7th May

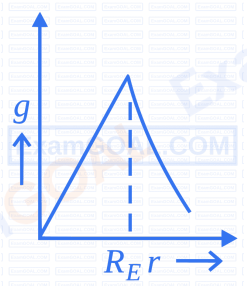
Which graph correctly presents the variation of acceleration due to gravity with the distance from the centre of the earth (radius of the earth $= R_E$)?

Open solution

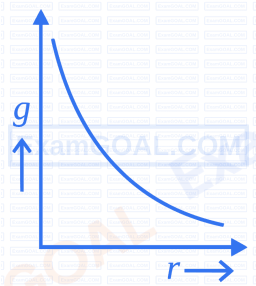
(A)



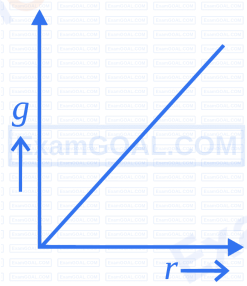
(B)



(C)



(D)



Q182. AIEEE 2012

The mass of a spaceship is $1000\ kg$. It is to be launched from the earth's surface out into free space. The value of g and R (radius of earth) are $10\ m/s^2$ and $6400\ km$ respectively. The required energy for this work will be:

Open solution

(A) 6.4×10^{11} Joules

(B) 6.4×10^{8} Joules

(C) 6.4×10^{9} Joules

(D) 6.4×10^{10} Joules

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Q183. AIEEE 2011

Two bodies of masses m and $4\,m$ are placed at a distance r. The gravitational potential at a point on the line joining them where the gravitational field is zero is:

(A) $-\frac{4Gm}{r}$ (B) $-\frac{6Gm}{r}$ (C) $-\frac{9Gm}{r}$ (D) zero

Open solution

Q184. AIEEE 2009

The height at which the acceleration due to gravity becomes $\frac{g}{9}$ (where $g =$ the acceleration due to gravity on the surface of the earth) in terms of R, the radius of the earth, is:

(A) $\frac{R}{\sqrt{2}}$ (B) $R/2$ (C) $\sqrt{2}\,R$ (D) $2\,R$

Open solution

Q185. AIEEE 2008

A planet in a distant solar system is 10 times more massive than the earth and its radius is 10 times smaller. Given that the escape velocity from the earth is $11\ km\,s^{-1}$, the escape velocity from the surface of the planet would be

(A) $1.1\ km\,s^{-1}$ (B) $100\ km\,s^{-1}$ (C) $110\ km\,s^{-1}$ (D) $0.11\ km\,s^{-1}$

Open solution

Q186. AIEEE 2008

This question contains Statement - 1 and Statement - 2. of the four choices given after the statements, choose the one that best describes the two statements.

Open solution

Statement - 1:

For a mass M kept at the center of a cube of side $'a'$, the flux of gravitational field passing through its sides $4\,\pi\,GM$.

Statement - 2:

If the direction of a field due to a point source is radial and its dependence on the distance $'r'$ from the source is given as $\frac{1}{r^2}$, its flux through a closed surface depends only on the strength of the source enclosed by the surface and not on the size or shape of the surface.

(A) Statement - 1 is false, Statement - 2 is true

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1

(C) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not a correct explanation for Statement - 1

(D) Statement - 1 is true, Statement - 2 is false

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Q187. AIEEE 2007

If g_E and g_M are the accelerations due to gravity on the surfaces of the earth and the moon respectively and if Millikan's oil drop experiment could be performed on the two surfaces, one will find the ratio $\frac{electronic\ charge\ on\ the\ moon}{electronic\ charge\ on\ the\ earth}$ to be

(A) g_M/g_E (B) 1 (C) 0 (D) g_E/g_M

Open solution

Q188. AIEEE 2005

The change in the value of g at a height h above the surface of the earth is the same as at a depth d below the surface of earth. When both d and h are much smaller than the radius of earth, then which one of the following is correct?

(A) $d = \frac{3h}{2}$ (B) $d = \frac{h}{2}$ (C) $d = h$ (D) $d = 2h$

Open solution

Q189. AIEEE 2005

A particle of mass $10\ g$ is kept on the surface of a uniform sphere of mass $100\ kg$ and radius $10\ cm$. Find the work to be done against the gravitational force between them to take the particle far away from the sphere (you may take $G = 6.67 \times 10^{-11}\ Nm^2/kg^2$)

(A) $3.33 \times 10^{-10}\ J$ (B) $13.34 \times 10^{-10}\ J$ (C) $6.67 \times 10^{-10}\ J$ (D) $6.67 \times 10^{-9}\ J$

Open solution

Q190. AIEEE 2005

Average density of the earth

(A) is a complex function of g (B) does not depend on g

(C) is inversely proportional to g (D) is directly proportional to g

Open solution

Q191. AIEEE 2004

A satellite of mass m revolves around the earth of radius R at a height x from its surface. If g is the acceleration due to gravity on the surface of the earth, the orbital speed of the satellite is

(A) $\frac{gR^2}{R+x}$ (B) $\frac{gR}{R-x}$

(C) gx (D) $\left(\frac{gR^2}{R+x}\right)^{1/2}$

Open solution

Q192. AIEEE 2004

The time period of an earth satellite in circular orbit is independent of

(A) both the mass and radius of the orbit

(B) radius of its orbit

(C) the mass of the satellite

(D) neither the mass of the satellite nor the radius of its orbit

Open solution

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Q193. AIEEE 2004

If g is the acceleration due to gravity on the earth's surface, the gain in the potential energy of an object of mass m raised from the surface of the earth to a height equal to the radius R of the earth is

(A) $\frac{1}{4}mgR$ (B) $2mgR$ (C) $\frac{1}{2}mgR$ (D) mgR

Open solution

Q194. AIEEE 2004

Suppose the gravitational force varies inversely as the nth power of distance. Then the time period of a planet in circular orbit of radius R around the sun will be proportional to

(A) R^n (B) $R^{(\frac{n-1}{2})}$
(C) $R^{(\frac{n+1}{2})}$ (D) $R^{(\frac{n-2}{2})}$

Open solution

Q195. AIEEE 2003

The time period of satellite of earth is 5 hours. If the separation between the earth and the satellite is increased to 4 times the previous value, the new time period will become

(A) 10 hours (B) 80 hours (C) 40 hours (D) 20 hours

Open solution

Q196. AIEEE 2003

Two spherical bodies of mass M and $5M$ & radii R & $2R$ respectively are released in free space with initial separation between their centers equal to $12R$. If they attract each other due to gravitational force only, then the distance covered by the smaller body just before collision is

(A) $2.5\ R$ (B) $4.5\ R$ (C) $7.5\ R$ (D) $1.5\ R$

Open solution

Q197. AIEEE 2003

The escape velocity for a body projected vertically upwards from the surface of earth is $11\ km/s$. If the body is projected at an angle of 45° with the vertical, the escape velocity will be

(A) $11\sqrt{2}\ km/s$ (B) $22\ km/s$ (C) $11\ km/s$ (D) $\frac{11}{\sqrt{2}}km/s$

Open solution

Q198. AIEEE 2002

The kinetic energy needed to project a body of mass m from the earth surface (radius R) to infinity is

(A) $mgR/2$ (B) $2mgR$ (C) mgR (D) $mgR/4$

Open solution

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Q199. AIEEE 2002

If suddenly the gravitational force of attraction between Earth and a satellite revolving around it becomes zero, then the satellite will

(A) continue to move in its orbit with same velocity

(B) move tangentially to the original orbit with the same velocity

(C) become stationary in its orbit

(D) move towards the earth

Open solution

Q200. AIEEE 2002

Energy required to move a body of mass m from an orbit of radius $2R$ to $3R$ is

(A) $\frac{GMm}{12R^2}$ (B) $\frac{GMm}{3R^2}$ (C) $\frac{GMm}{8R}$ (D) $\frac{GMm}{6R}$

Open solution

Q201. AIEEE 2002

The escape velocity of a body depends upon mass as

(A) m^0 (B) m^1 (C) m^2 (D) m^3

Open solution

Numerical 16 questions

Q202. JEE Main 2025 (Online) 3rd April Morning Shift

Three identical spheres of mass m , are placed at the vertices of an equilateral triangle of length a. When released, they interact only through gravitational force and collide after a time $\mathrm{T} = 4$ seconds. If the sides of the triangle are increased to length $2a$ and also the masses of the spheres are made 2 m , then they will collide after_____seconds.

Open solution

Q203. JEE Main 2025 (Online) 2nd April Evening Shift

A satellite of mass 1000 kg is launched to revolve around the earth in an orbit at a height of 270 km from the earth's surface. Kinetic energy of the satellite in this orbit is_____ $\times 10^{10}$ J.

(Mass of earth $= 6 \times 10^{24}$ kg, Radius of earth $= 6.4 \times 10^{6}$ m, Gravitational constant $= 6.67 \times 10^{-11} \mathrm{Nm}^2\,\mathrm{kg}^{-2}$)

Open solution

Q204. JEE Main 2025 (Online) 29th January Evening Shift

Two planets, A and B are orbiting a common star in circular orbits of radii R_A and R_B, respectively, with $R_B = 2R_A$. The planet B is $4\sqrt{2}$ times more massive than planet A. The ratio $\left(\frac{\mathrm{L_B}}{\mathrm{L_A}}\right)$ of angular momentum (L_B) of planet B to that of planet A (L_A) is closest to integer _____.

Open solution

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Q205. JEE Main 2025 (Online) 24th January Evening Shift

Acceleration due to gravity on the surface of earth is ' g '. If the diameter of earth is reduced to one third of its original value and mass remains unchanged, then the acceleration due to gravity on the surface of the earth is _____ g.

Open solution

Q206. JEE Main 2025 (Online) 23rd January Evening Shift

A satellite of mass $\frac{M}{2}$ is revolving around earth in a circular orbit at a height of $\frac{R}{3}$ from earth surface. The angular momentum of the satellite is $\mathrm{M}\sqrt{\frac{\mathrm{GMR}}{x}}$. The value of x is _____ , where M and R are the mass and radius of earth, respectively. (G is the gravitational constant)

Open solution

Q207. JEE Main 2024 (Online) 30th January Evening Shift

A simple pendulum is placed at a place where its distance from the earth's surface is equal to the radius of the earth. If the length of the string is $4m$, then the time period of small oscillations will be _____ s. [take $g = \pi^2 ms^{-2}$]

Open solution

Q208. JEE Main 2022 (Online) 29th July Morning Shift

If the acceleration due to gravity experienced by a point mass at a height h above the surface of earth is same as that of the acceleration due to gravity at a depth $\alpha \mathrm{h}\,(\mathrm{h} << \mathrm{R_e})$ from the earth surface. The value of α will be _____.

(use $\mathrm{R_e} = 6400\ \mathrm{km}$)

Open solution

Q209. JEE Main 2022 (Online) 25th June Evening Shift

Two satellites S_1 and S_2 are revolving in circular orbits around a planet with radius R_1 = 3200 km and R_2 = 800 km respectively. The ratio of speed of satellite S_1 to be speed of satellite S_2 in their respective orbits would be $\frac{1}{x}$ where x = _____.

Open solution

Q210. JEE Main 2021 (Online) 1st September Evening Shift

Two satellites revolve around a planet in coplanar circular orbits in anticlockwise direction. Their period of revolutions are 1 hour and 8 hours respectively. The radius of the orbit of nearer satellite is 2×10^3 km. The angular speed of the farther satellite as observed from the nearer satellite at the instant when both the satellites are closest is $\frac{\pi}{x} rad\, h^{-1}$ where x is _____.

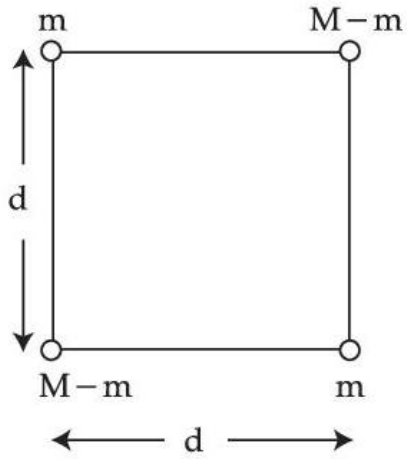
Open solution

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Q211. JEE Main 2021 (Online) 27th August Morning Shift

A body of mass (2M) splits into four masses (m, M − m, m, M − m}, which are rearranged to form a square as shown in the figure. The ratio of $\frac{M}{m}$ for which, the gravitational potential energy of the system becomes maximum is x : 1. The value of x is ..... .

Open solution

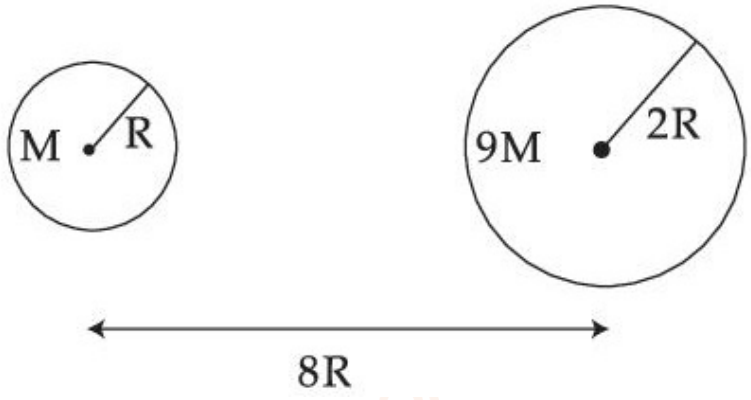


Q212. JEE Main 2021 (Online) 27th July Morning Shift

Suppose two planets (spherical in shape) in radii R and 2R, but mass M and 9M respectively have a centre to centre separation 8 R as shown in the figure. A satellite of mass 'm' is projected from the surface of the planet of mass 'M' directly towards the centre of the second planet. The minimum speed 'v' required for the satellite to reach the surface of the second planet is $\sqrt{\frac{a}{7}\frac{GM}{R}}$ then the value of 'a' is _____.

Open solution

[Given : The two planets are fixed in their position]



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Q213. JEE Main 2021 (Online) 17th March Morning Shift

The radius in kilometer to which the present radius of earth (R = 6400 km) to be compressed so that the escape velocity is increased 10 times is _____.

Open solution

Q214. JEE Main 2021 (Online) 16th March Evening Shift

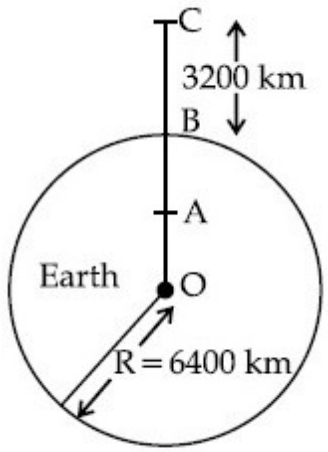
If one wants to remove all the mass of the earth to infinity in order to break it up completely.

The amount of energy that needs to be supplied will be $\frac{x}{5}\frac{GM^2}{R}$ where x is _____ (Round off to the Nearest Integer) (M is the mass of earth, R is the radius of earth, G is the gravitational constant)

Open solution

Q215. JEE Main 2021 (Online) 26th February Evening Shift

In the reported figure of earth, the value of acceleration due to gravity is same at point A and C but it is smaller than that of its value at point B (surface of the earth). The value of OA : AB will be x : y. The value of x is _____.



Open solution

Q216. JEE Main 2021 (Online) 25th February Evening Shift

The initial velocity v_i required to project a body vertically upward from the surface of the earth to reach a height of 10R, where R is the radius of the earth, may be described in terms of escape velocity v_e such that $v_i = \sqrt{\frac{x}{y}} \times v_e$. The value of x will be _____.

Open solution

Q217. JEE Main 2020 (Online) 8th January Evening Slot

An asteroid is moving directly towards the centre of the earth. When at a distance of 10R (R is the radius of the earth) from the earths centre, it has a speed of 12 km/s. Neglecting the effect of earths atmosphere, what will be the speed of the asteroid when it hits the surface of the earth (escape velocity from the earth is 11.2 km/s) ? Give your answer to the nearest integer in kilometer/s _____.

Open solution

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MCQ (More than One Correct Answer) 1 questions

Q218. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

An astronaut of mass m is working on a satellite orbiting the earth at a distance h from the earth's surface. The radius of the earth is R, while its mass is M. The gravitational pull F_G on the astronaut is :

(A) Zero since astronaut feels weightless

(B) $0 < F_G < \frac{GMm}{R^2}$

(C) $\frac{GMm}{(R+h)^2} < F_G < \frac{GMm}{R^2}$

(D) $F_G = \frac{GMm}{(R+h)^2}$

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **B**	3. **C**	4. **A**	5. **B**
6. **B**	7. **C**	8. **D**	9. **A**	10. **B**
11. **C**	12. **C**	13. **A**	14. **C**	15. **C**
16. **D**	17. **B**	18. **B**	19. **B**	20. **A**
21. **D**	22. **D**	23. **C**	24. **D**	25. **A**
26. **C**	27. **A**	28. **D**	29. **C**	30. **A**
31. **D**	32. **A**	33. **A**	34. **A**	35. **D**
36. **C**	37. **C**	38. **D**	39. **D**	40. **D**
41. **B**	42. **D**	43. **C**	44. **D**	45. **D**
46. **D**	47. **A**	48. **D**	49. **A**	50. **B**
51. **B**	52. **A**	53. **A**	54. **D**	55. **A**
56. **B**	57. **B**	58. **A**	59. **C**	60. **D**
61. **C**	62. **C**	63. **D**	64. **C**	65. **D**
66. **B**	67. **B**	68. **B**	69. **A**	70. **C**
71. **D**	72. **B**	73. **D**	74. **D**	75. **A**
76. **D**	77. **C**	78. **A**	79. **D**	80. **B**
81. **B**	82. **A**	83. **A**	84. **D**	85. **A**
86. **B**	87. **D**	88. **B**	89. **A**	90. **C**
91. **C**	92. **C**	93. **A**	94. **A**	95. **D**
96. **A**	97. **D**	98. **D**	99. **B**	100. **B**
101. **D**	102. **B**	103. **D**	104. **A**	105. **B**
106. **D**	107. **C**	108. **D**	109. **D**	110. **D**
111. **B**	112. **B**	113. **A**	114. **B**	115. **B**
116. **A**	117. **A**	118. **B**	119. **C**	120. **C**
121. **A**	122. **C**	123. **C**	124. **D**	125. **B**
126. **D**	127. **C**	128. **C**	129. **A**	130. **A**
131. **D**	132. **C**	133. **D**	134. **B**	135. **B**
136. **A**	137. **C**	138. **A**	139. **B**	140. **C**
141. **C**	142. **B**	143. **D**	144. **D**	145. **D**
146. **A**	147. **B**	148. **B**	149. **D**	150. **D**

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151. **B**	152. **A**	153. **D**	154. **A**	155. **B**
156. **C**	157. **D**	158. **D**	159. **D**	160. **A**
161. **B**	162. **C**	163. **A**	164. **C**	165. **D**
166. **B**	167. **C**	168. **D**	169. **D**	170. **B**
171. **A**	172. **C**	173. **D**	174. **B**	175. **A**
176. **C**	177. **A**	178. **C**	179. **B**	180. **A**
181. **B**	182. **D**	183. **C**	184. **D**	185. **C**
186. **B**	187. **B**	188. **D**	189. **C**	190. **D**
191. **D**	192. **C**	193. **C**	194. **C**	195. **C**
196. **C**	197. **C**	198. **C**	199. **B**	200. **D**
201. **A**				

Numerical

202. **8**	203. **3**	204. **8**	205. **9**	206. **3**
207. **8**	208. **2**	209. **2**	210. **3**	211. **2**
212. **4**	213. **64**	214. **3**	215. **4**	216. **10**
217. **16**				

MCQ (More than One Correct Answer)

218. **B, D**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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